

CHAPTER 20 Science inquiry skills in chemistry

This chapter covers the skills needed to successfully plan, conduct and evaluate results from experiments and investigations in chemistry.

For investigations, you will learn how to develop inquiry questions and hypotheses. There are many aspects to designing an investigation and you will consider risk assessments and ethical concerns, and the identification, measurement and control of variables which allow for reliable collection of data. Uncertainty and error are common to all experimental work, and you will learn to identify causes of random and systematic error, and importantly, how appropriate materials, technology and procedures can be used to accurately collect and record data to reduce uncertainty in experimental data.

The chapter describes how best to represent your data, how to identify trends and patterns and explains how to analyse your results, including using mathematical models. Many investigations in chemistry involve solving problems, such as the identification of an unknown or a determination of a quantity. Strategies for utilising critical thinking that will assist in problem-solving tasks are discussed.

Finally, the sharing of findings is vital, and in this chapter you will learn how to effectively communicate conclusions using appropriate scientific language, nomenclature and scientific notation.

Science inquiry skills

- identify, research and refine questions for investigation; propose hypotheses; and predict possible outcomes
- design investigations, including the procedure(s) to be followed, the materials required, and the type and amount of primary and/or secondary data to be collected; conduct risk assessments; and consider research ethics
- conduct investigations safely, competently and methodically for the collection of valid and reliable data, including: the use of devices to accurately measure temperature change and mass, flame tests, separation techniques and heat of reaction
- conduct investigations safely, competently and methodically for the collection of valid and reliable data, including: chromatography, measuring pH, rate of reaction, identification of the products of reactions, and determination of solubilities of ionic compounds to recognise patterns in solubility
- represent data in meaningful and useful ways, including using appropriate graphic representations and correct units and symbols; organise and process data to identify trends, patterns and relationships; identify sources of random and systematic error and estimate their effect on measurement results; and select, synthesise and use evidence to make and justify conclusions
- interpret a range of scientific and media texts, and evaluate processes, claims and conclusions by considering the quality of available evidence; and use reasoning to construct scientific arguments
- communicate to specific audiences and for specific purposes using appropriate language, nomenclature and formats, including scientific reports

FPO
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20.1 Questioning

TYPES OF INVESTIGATIONS

There are many different types of investigation that can be conducted in chemistry. You are probably most familiar with practical investigations or experiments. Other types of investigations involve researching and evaluating primary and/or secondary sources of information to answer an inquiry question. Examples of the types of different investigations are listed in Table 20.1.1 and Table 20.1.2.

TABLE 20.1.1 Examples of primary research

Primary research	Example tasks
conducting experiments in a laboratory	- planning a valid experiment - conducting a risk assessment - working safely - recording observations and results - analysing and evaluating data and information
conducting field work	- conducting a risk assessment - working safely - recording observations and results - analysing and evaluating data and information
conducting surveys	- writing questions - collecting data - analysing data and information
designing a model	- identifying a problem to be modelled - summarise key findings - identify advantages and limitations of the model

TABLE 20.1.2 Examples of secondary research

Secondary research	Example tasks
researching published data from primary and secondary sources	- finding published information in scientific magazines and journals, books, databases, media texts, labels of commercially available products - analysing and evaluating data and information

Before you are able to start an investigation, you first need to understand the theory behind it. This section is a guide to some of the key steps that should be taken when first developing inquiry questions and hypotheses.

DEVELOPING AIMS AND QUESTIONS, FORMULATING HYPOTHESES AND MAKING PREDICTIONS

The inquiry question, aim and hypothesis are interlinked (Figure 20.1.1). It is important to note that each of these can be refined as the planning of the investigation continues.

Inquiry questions and hypotheses

An inquiry question defines what is being investigated, for example: ‘What is the relationship between concentration and rate of reaction?’

A **hypothesis** is a prediction based on previous knowledge and evidence or observations that attempts to answer the inquiry question. For example: Increasing the concentration of the acid will increase the rate of the reaction of hydrochloric acid solution with marble chips.



FIGURE 20.1.1 There may be many elements to an investigation. Taking a step-by-step approach will help the process and assist in completing a solid and worthwhile investigation.

It is important that you can interpret what an inquiry question is asking you to do. To do this:

- identify a ‘guiding’ word, such as *who, what, where, why*
- link the guiding word to command verbs that are often used in senior high school such as *identify, describe, compare, contrast, distinguish, analyse, evaluate* and *create*.

Table 20.1.3 gives examples of inquiry questions that could be investigated.

TABLE 20.1.3 How to interpret an inquiry question

Guiding word	Example inquiry questions	What are you being asked to do? What are the command verbs?
What	What difference can nanomaterials make to society and the environment? What are electrons, protons and neutrons made of?	Identify and describe specific examples, evidence, reasons and analogies from a variety of possibilities. <i>Identify</i> and <i>describe</i> .
Where	Where would an element with an atomic number of 130 be placed in the modern periodic table, what properties would it have and how likely is it to be discovered?	Identify and describe giving reasons for a place or location. <i>Identify</i> and <i>describe</i> .
How	How are atoms ‘seen’? How can metal ores be transformed into metals? How do different crude oil extraction methods compare in terms of their ease of extraction and environmental impact?	Identify and describe in detail a process or mechanism. Give examples using evidence and reasons. <i>Identify</i> and <i>describe</i> .
Why	Why are the ten most abundant elements in the universe not the same as the ten most abundant elements on Earth? Why does the composition of crude oil vary between different oil wells?	Explain in detail the causes, reasons, mechanisms and evidence for. <i>Identify</i> and <i>describe</i> .
Would	Would there be life if elements did not form compounds?	Evaluate. Justify, giving reasons for and against (using evidence, analogies, comparisons). <i>Evaluate, assess, justify</i> .
Is/are	Are there more elements to be discovered? Is it an advantage or a disadvantage to speed up chemical reactions used in cooking?	Evaluate. Justify, giving reasons and evidence. <i>Evaluate, assess, justify</i> .
On what basis	On what basis are alternative forms of the periodic table constructed?	Evaluate. Justify using reasons and evidence. <i>Evaluate, assess, justify</i> .
Can	Can we live without the lanthanide and actinides elements?	Evaluate and assess. Is it possible? Give reasons, suggest possible alternatives. <i>Evaluate, assess, justify, create</i> .
Do/does	Do the lanthanide and actinides rust or corrode? Do we need crude oil?	Evaluate. Justify using reasons and evidence for and against. <i>Evaluate, assess, justify</i> .
Should	Should cars be made from shape memory metals?	Evaluate pros and cons, implications and limitations. Make a judgement. <i>Evaluate, assess, justify, create</i> .
Might	What might we do if crude oil supplies become too expensive?	Evaluate. Justify, giving reasons for and against (using evidence, analogies, comparisons). <i>Evaluate, assess, justify, compare, contrast, create</i> .

Formulating a question

If you are required to formulate a question, it is good practice to research the topic to be investigated. You should become familiar with the relevant scientific concepts and key terms.

During this research, write down questions or correlations as they arise.

Compile a list of possible ideas. Do not reject ideas that initially might seem improbable. Use these ideas to generate questions that are answerable.

Before constructing a hypothesis, formulate a question that needs an answer. This question will lead to a hypothesis when:

- the question is confined to measurable variables
- a prediction is made based on knowledge and experience.

Evaluating your question

Once a question has been chosen, stop to evaluate the question before progressing. The question may need further refinement or even further investigation before it is suitable as a basis for investigation. A major planning point is to not attempt something that is not possible to complete in the time available or with the resources on hand. For example, it might be difficult to create a specific piece of specialist equipment in the school laboratory or if experiments take too long it may be impossible to conduct sufficient trials in the time allocated.

To evaluate the question, consider the following:

- **Relevance:** Is the question related to the appropriate course content?
- **Clarity and measurability:** Can the question be framed as a clear hypothesis?

If the question cannot be stated as a specific hypothesis, then it is going to be very difficult to complete the investigation.

- **Time frame:** Can the question be answered within a reasonable period of time? Is the question too broad?
- **Knowledge and skills:** Do you have a level of knowledge and the laboratory skills that will allow the question to be explored?
- **Practicality:** Are resources, such as laboratory equipment and materials, likely to be readily available?
- **Safety and ethics:** Consider the safety and ethical issues associated with the question you will be investigating. If there are issues, can they be addressed?

Sourcing information

Once you have selected a topic, the next step is to source reliable information. Some of the steps involved in sourcing information are:

- identifying key terms
- sourcing information
- evaluating the credibility of sources
- evaluating experimental data/evidence.

Sources can be:

- **primary sources**—original sources of data and evidence; for example, articles containing research findings that have been published in peer-reviewed scientific journals or research presented at a scientific conference.
- **secondary sources**—analyses and interpretations of primary sources; for example, textbooks, magazine articles and newspaper articles.

Some of the sources that may contain useful information are listed in Table 20.1.4.

TABLE 20.1.4 Useful sources of primary and secondary information

Primary sources of information	Secondary sources of information
journal articles from peer-reviewed journals, such as <i>Nature</i>	newspaper articles and opinion pieces
global databases, statistics and surveys	magazine articles; examples of reputable science magazines include: • <i>Cosmos</i> • <i>Double Helix</i> (CSIRO) • <i>New Scientist</i> • <i>Popular Science</i> • <i>Scientific American</i>
laboratory work	government reports
computer simulations and modelling	

Hypothesis

A hypothesis is a prediction that is based on evidence and prior knowledge. A hypothesis often takes the form of a proposed relationship between two or more variables in a cause and effect relationship; or in other words, 'If X is done, then Y will occur.'

Here are some examples of hypotheses:

- If the number of carbon atoms per molecule of a hydrocarbon fuel increases, the energy released per kilogram of fuel during combustion will increase.
- If the concentration of sulfuric acid in a reaction with solid copper(II) oxide is increased, the rate of reaction will increase proportionally to the increase in concentration.
- If the structure of hydrocarbons with the same molecular formula varies (isomers), the boiling point of the isomers will also vary.

Variables

An investigation can test a good scientific hypothesis and the hypothesis will be supported or refuted. To be a testable hypothesis, it should be possible to measure both what is changed or carried out and what will happen. The factors that are monitored during an experiment or investigation are called the **variables**. An experiment or investigation determines the relationship between variables by measuring the results.

There are three categories of variables:

- The **independent variable** is the variable that is determined by the researcher (the one that is selected and changed).
- The **dependent variable** is the variable that may change in response to a change in the independent variable. This is the variable that will be measured or observed.
- **Controlled variables** are all the variables that must be kept constant during the investigation.

Note that you should only test one variable at a time, otherwise it cannot be stated that the changes in the dependent variable are the result of changes in the independent variable.

Qualitative and quantitative variables

Variables are either qualitative or quantitative, with further subsets in each category.

- **Qualitative** variables can be observed but not measured. They can only be sorted into groups or categories such as brightness, type of material, colour of flame.
- **Quantitative** variables can be measured. Mass, volume, temperature, pH and time are all examples of quantitative data.
 - Discrete variables consist of only integer numerical values, not fractions; for example, the number of protons in an atom, the number of atoms of each element in a compound and the number of isotopes of a particular element.
 - Continuous variables allow for any numerical value within a given range; for example, the measurement of temperature, volume, mass, pH and conductivity.

Formulating a hypothesis

Once the inquiry question is confirmed, formulating a hypothesis comes next. A hypothesis requires a proposed relationship between two variables. It should predict that a relationship exists or does not exist.

Identify the two variables in your question. State the independent and dependent variables.

For example: If I do/change this (independent variable), then this (dependent variable) will happen.

A good hypothesis should:

- be a statement
- be based on information contained in the research question (purpose)
- be worded so that it can be tested in the experiment
- include an independent and a dependent variable
- include variables that are measurable.

The hypothesis should also be falsifiable. This means that a negative outcome would disprove it. For example, the hypothesis that all apples are round cannot be proved beyond doubt, but it can be disproved—in other words, it is falsifiable. In fact, only one square apple is needed to disprove this hypothesis. Unfalsifiable hypotheses cannot be proved by science. These include hypotheses on ethical, moral and other subjective judgements.

Worked example 20.1.1

DEVELOPING A HYPOTHESIS

Develop a hypothesis and identify variables for investigating the rate of the reaction between marble chips and hydrochloric acid.	
Thinking	Working
Write an inquiry question.	What is the relationship between the concentration of the acid and reaction rate?
Identify the independent variable.	concentration of hydrochloric acid solution
Identify the dependent variable.	the mass lost during the first 30 seconds of the reaction
Identify the controlled variables.	starting temperature of hydrochloric acid volume of hydrochloric acid mass of the marble chips surface area of the marble chips
Write a potential hypothesis.	For a constant particle size of marble chips and volume of hydrochloric acid, if the concentration of the hydrochloric acid solution is halved, the rate of reaction will halve.

Worked example: Try yourself 20.1.1

DEVELOPING A HYPOTHESIS

- In this example, the heat given off in the combustion reactions of a range of hydrocarbons is measured by burning a certain amount of hydrocarbon and measuring the heat produced by measuring the temperature change of water. A possible set-up for the experiment is shown in Figure 20.1.2.

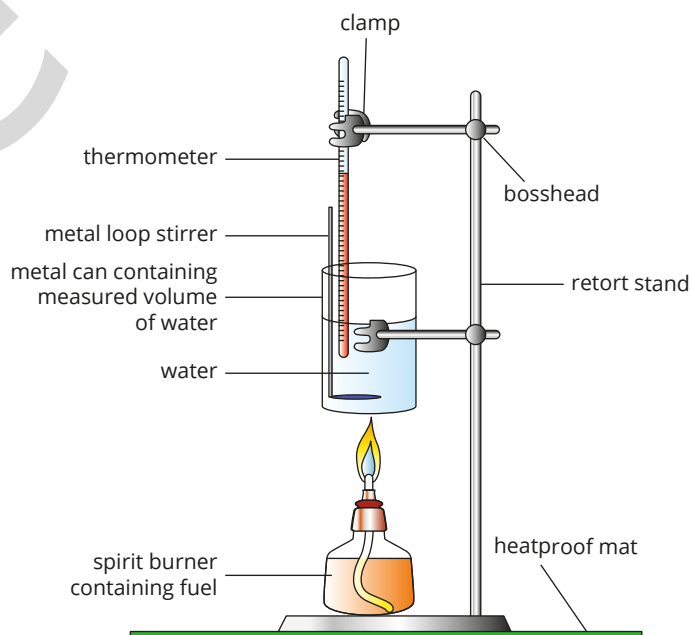


FIGURE 20.1.2 Apparatus for measuring heat energy released when a fuel burns

Develop a hypothesis and identify variables for investigating how the molecular mass of a hydrocarbon affects the energy released per gram of the hydrocarbon.

20.1 Review

SUMMARY

- By utilising data from primary and/or secondary sources, you will better understand the context of your investigation to create an informed inquiry question.
- A hypothesis is a prediction based on previous knowledge and evidence or observations that attempts to answer the research question. For example: 'If the temperature of the acid is increased, the rate of reaction of calcium carbonate and the hydrochloric acid will increase.'
- Once a question has been chosen, evaluate the question before progressing. The question may need further refinement or even further investigation before it is suitable as a basis for an achievable and worthwhile investigation.
- There are three categories of variables.
 - The independent variable is the variable that is controlled by the researcher.
 - The dependent variable is the variable that may change in response to a change in the independent variable. This is the variable that will be measured or observed.
 - Controlled variables are all the variables that must be kept constant during the investigation.

KEY QUESTIONS

- 1 Scientists make observations from which a hypothesis is stated and this is then experimentally tested.
 - a Define 'hypothesis'.
 - b How are theories and principles different from a hypothesis?
- 2 Which of the following is an example of an inquiry question?
 - A How are chemicals in solutions measured?
 - B A compound consists of two or more elements.
 - C Decreasing the volume of a container of gas will increase the pressure.
 - D The mass of the reactants equalled the mass of the products.
- 3 For each of the following hypotheses, select the dependent variable.
 - a If water is filtered through a domestic water purifier, then the electrical conductivity of the water will decrease.
 - b The concentration of lead in water will be higher in storm water close to an industrial site than in drinking water.
 - c The electrical conductivity of water from Fremantle Harbour will be greatest where ocean water can mix with fresh water.
 - d The pH of commercially available sparkling mineral water will be lower than commercially available non-sparkling mineral water.
- 4 In describing an experiment a student uses the following range of values to record flame tests of ionic compounds: yellow, lilac, red and green. What type of measurement is the variable 'colour'?
- 5 Which of the following methods is likely to be the most accurate quantitative method for measuring the pH of water?
 - A using pH paper (e.g. litmus paper)
 - B using universal indicator and a colour chart
 - C using a calibrated pH meter at a particular temperature
 - D using a conductivity meter
- 6 Select the best hypothesis from the three options below. Give two reasons for your choice.
 - A Hypothesis 1: If volume and temperature of a gas are changed, then this will affect the pressure of the gas.
 - B Hypothesis 2: Concentration of solutions can be expressed using different units.
 - C Hypothesis 3: If water is filtered through a domestic water purifier, then the electrical conductivity of the water will decrease.
- 7 Give the correct term that describes an experiment with each of the following conditions.
 - a The experiment addresses the hypothesis and aims.
 - b The experiment is repeated and consistent results are obtained.
 - c Appropriate equipment is chosen for the desired measurements.

20.2 Planning investigations

Once you have formulated your hypothesis, you will need to plan and design your investigation. Taking the time to carefully plan and design a practical investigation before beginning will help you to maintain a clear and concise focus throughout. Preparation is essential. This section is a guide to some of the key steps that should be taken when planning and designing an investigation.

DEVELOPING THE METHOD

The method of your investigation is a step-by-step procedure. When detailing the method, ensure it complies as a valid, reliable and accurate investigation.

Validity

Validity refers to whether an experiment or investigation is in fact testing the set hypothesis and aims. Is the investigation obtaining data that is relevant to the question?

To ensure an investigation is valid, it should be designed so that only one variable is being changed at a time. The remaining variables must remain constant so that meaningful conclusions can be drawn about the effect of each variable in turn.

To ensure validity, carefully determine:

- the independent variable: the variable that will be changed, and how it will change
- the dependent variable: the variable that will be measured
- the controlled variables: the variables that must remain constant, and how they will be maintained.

Reliability

Reliability refers to the notion that if the experiment is repeated many times, the results obtained should be consistent.

It is important to determine how many times the experiment needs to be replicated (Figure 20.2.1). Many scientific investigations lack sufficient repetition to ensure that the results can be considered reliable and repeatable.

- Repeat readings: repeat each reading (at least three times if possible), record each measurement and then average the measurements. This reduces systematic errors and allows random errors to be identified. If a reading differs too much from the rest (known as an **outlier**), discard it before averaging.
- Sample size: where there might be differences in construction or manufacture of a sample, there should be various samples with the same conditions in the same experiment. The greater the sample size the more reliable the data.
- Repeats: if possible repeat the experiment on a different day. Don't change anything. If the results are not the same, think about what could have happened. For example, was the equipment faulty, or were all the controlled variables correctly identified? Repeat the experiment a third time to confirm which run was correct. Repeat an experiment as many times as possible; three is a good number but, if time and resources allow, aim for five.

In some situations, it may be appropriate to conduct a **control experiment**. The control is an identical experiment carried out at the same time, except that in the control experiment the independent variable is not changed.

When the controls do not behave as expected, the data obtained from an experiment or observation is not reliable.

Accuracy and precision

In science and statistics, the terms 'accuracy' and 'precision' have very specific and different meanings:

- **Accuracy** is the ability to obtain the correct measurement. To obtain accurate results, you must minimise systematic errors.

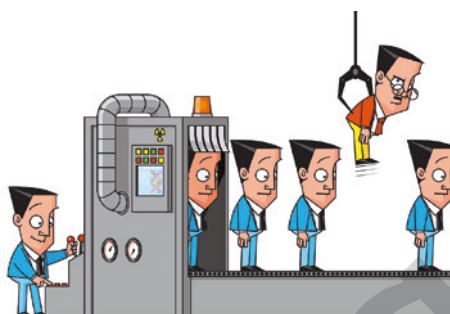


FIGURE 20.2.1 Replication increases the reliability of your investigation. It ensures that if anyone repeats the investigation they will obtain similar data.

- **Precision** is the ability to consistently obtain the same measurement. To obtain precise results, you must minimise random errors. You will learn more about types of errors in Section 20.3.

To understand more clearly the difference between accuracy and precision, think about firing arrows at an archery target (Figure 20.2.2). Accuracy is being able to hit the bullseye, whereas precision is being able to hit the same spot every time you shoot. If you hit the bullseye every time you shoot, you are both accurate and precise (Figure 20.2.2a). If you hit the same area of the target every time but not the bullseye, you are precise but not accurate (Figure 20.2.2b). If you hit the area around the bullseye each time but don't always hit the bullseye, you are accurate but not precise (Figure 20.2.2c). If you hit a different part of the target every time you shoot, you are neither accurate nor precise (Figure 20.2.2d).

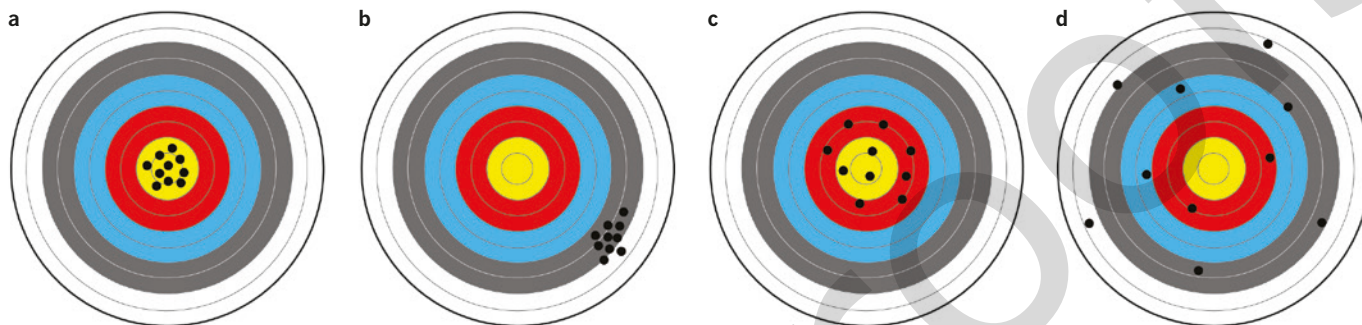


FIGURE 20.2.2 Examples of accuracy and precision: (a) both accurate and precise, (b) precise but not accurate, (c) accurate but not precise, and (d) neither accurate nor precise

Applying this idea to a chemistry situation, if you are timing the rate of reaction by stopping a stopwatch when 100 mL of gas has been released, and the you get consistent results but you are always stopping the stopwatch too early, your results are precise but not accurate.

Uncertainty

When scientists take a measurement, the measurement is always subject to variations in accuracy and precision. The goal when taking measurements is to get as close as possible to 'true' or 'correct' value. The **uncertainty** is how far an experimental quantity might be from the true value. It can be calculated, or estimated and is often represented as a percentage. For example, if the uncertainty was described as $\pm 1\%$, it means that measurement is likely to be no more than 1% above or below the 'true' value of the quantity that is being measured.

In this course you are not required to calculate uncertainties, but it is a good idea to consider uncertainty when planning experiments, and later when you come to evaluate your results.

Recording numerical data

Reasonable steps to ensure the accuracy of the investigation include considering:

- the unit in which the independent and dependent variables will be measured
- the instrument that will be used to measure the independent and dependent variables.

Select and use appropriate equipment, materials and methods. For example, select equipment that measures to greater precision to reduce uncertainty and repeat the measurements to confirm them.

When using measuring instruments, the number of significant figures and decimal places you use is determined by how precise your measurements are.

This depends on the scale, accuracy and precision of the instrument and technique you are using (Figure 20.2.3). For example, a beaker is only used to measure volumes approximately and has a relatively high uncertainty, for example $\pm 5\%$. A graduated pipette is more accurate, with uncertainties of $\pm 0.1\%$ or $\pm 0.2\%$. Your pipette may be accurate but if your technique using the pipette is variable, the overall accuracy and precision will be limited.

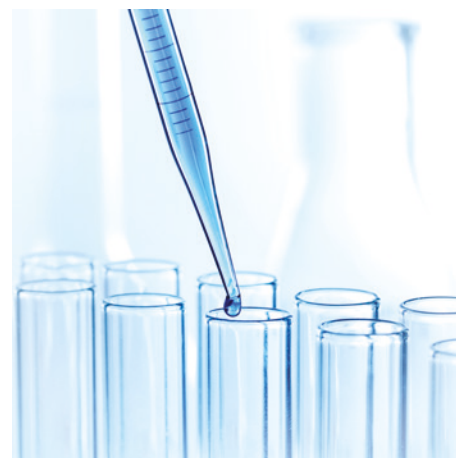


FIGURE 20.2.3 A 5 mL graduated pipette can measure volumes to an accuracy of one hundredth of a millilitre, or $5 \text{ mL} \pm 0.01 \text{ mL}$. The pipette has major divisions of 1 mL and minor divisions of 0.1 mL. You can estimate to 0.01 mL and record volumes to 2 decimal places, for example 3.80 mL or 4.52 mL.

When you record **raw data** and report processed data, use the number of significant figures available from your equipment or observation. Using either a greater or smaller number of significant figures can be misleading. For example, Table 20.2.1 shows measurements samples of a chemical taken using an electronic balance accurate to two decimal places. The data was entered into a spreadsheet to calculate the mean, which was displayed with four decimal places. You would record the mean as 20.83 g, not 20.8260 g, because two decimal places is the precision limit of the instrument. Recording 20.8260 g would be an example of false precision.

TABLE 20.2.1 An example of false precision in a data calculation

Sample	1	2	3	4	5	Mean
Mass (g)	20.13	20.62	21.22	20.99	21.17	20.8260

Data analysis

Data analysis is part of the method. Consider how the data will be presented and analysed. A wide range of analysis tools are available. For example, tables organise data so that patterns can be established and graphs can show relationships and comparisons. In fact, preparing an empty table showing the data that needs to be obtained will help in the planning of the investigation.

Sourcing appropriate equipment and materials

When designing your investigation, you will need to decide on the materials, technology and instrumentation that will be used to carry out your investigation. It is important to find the right balance between items that are easily accessible and those which will give you accurate results. When conducting your investigation, it will be important to take note of the precision of your chosen instrumentation and how this affects the accuracy and validity of your results.

Modifying the method

The method may need modifying as the investigation is carried out. The following actions will help to determine any issues in the method and how to modify them:

- Record everything.
- Be prepared to make changes to the approach.
- Note any difficulties encountered and the ways they were overcome. What were the failures and successes? Every test carried out can contribute to the understanding of the investigation as a whole, no matter how much of a disaster it may first appear.

If the expected data is not obtained, don't worry. As long as it can be critically and objectively evaluated, the limitations of the investigation are identified and further investigations proposed, the work is worthwhile.

COMPLYING WITH ETHICAL AND SAFETY GUIDELINES

Ethical considerations

When planning an investigation, researchers should always identify all possible ethical considerations and evaluate their necessity or ways that can reduce or mitigate them.

Ethical issues might include:

- How can this affect wider society?
- Does one party benefit over another; for example, one individual, a group of individuals or a community?
- Is there a risk of harm (physical or mental) to people involved in the research?
- Does it prevent anyone from gaining their basic needs?
- How can this impact on future ethical decisions or issues?

- Does the research cause damage to the environment?
- Does the research cause harm to other living things?

In reality, school chemistry investigations generally will have minor ethical issues, if any, but these should be considered in your planning.

Safety in practical experiments and investigations

While planning for an investigation, it is important for your safety and the safety of others that the potential risks are considered (Figure 20.2.4).

Everything we do has some risk involved. Risk assessments are performed to identify, assess and control hazards. Always identify the risks and control them to keep everyone safe.



FIGURE 20.2.4 When planning an investigation you need to identify, assess and control hazards.

To identify risks think about:

- the activity that will be carried out
- the equipment or chemicals that will be used.

The following hierarchy of risk controls is organised from the most effective to least effective:

- 1 *Elimination*: Eliminate dangerous equipment, procedures or substances.
- 2 *Substitution*: Find different equipment, procedures or substances to use that will achieve the same result, but have less risk associated.
- 3 *Isolation*: Ensure there is a barrier between the person and the hazard. Examples include physical barriers such as guards in machines, or fume hoods to work with volatile substances.
- 4 *Administrative controls*: Provide guidelines, special procedures, warning signs and safe behaviours for any participants.
- 5 *Personal protective equipment (PPE)*: Wear safety glasses, lab coats, gloves and respirators etc. where appropriate, and provide these to other participants.

Figure 20.2.5 shows a flow chart of how to consider and assess the risks involved in a research investigation.

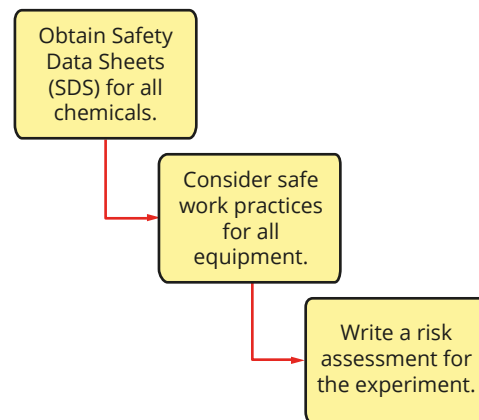


FIGURE 20.2.5 Steps involved in identifying risks

Chemical codes

The chemicals at school or at the hardware shop have a warning symbol on the label. These are **chemical (HAZCHEM) codes**. Some common codes and their meanings are shown in Figure 20.2.6. Trucks that carry chemicals display hazard symbols, as shown in Figure 20.2.7.

HAZCHEM INTERPRETATION

NUMBER		
1		Water Jets
2		Water Fog
3		Foam
4		Dry Agent
FIRST LETTER		
P	V	Full Protective Clothing*
R		Full Protective Clothing*
S	V	Breathing Apparatus
S	V	Breathing Apparatus for Fire Only
T		Breathing Apparatus
T		Breathing Apparatus for Fire Only
W	V	Full Protective Clothing*
X		Full Protective Clothing*
Y	V	Breathing Apparatus
Y	V	Breathing Apparatus for Fire Only
Z		Breathing Apparatus
Z		Breathing Apparatus for Fire Only
SECOND LETTER		
E		Consider Evacuation

Note V: Danger of violent reaction or explosion

* Full Protective Clothing includes Breathing Apparatus

1	EXPLOSIVE (Gunpowder, flares)		4.3	DANGEROUS WHEN WET (Calcium carbide)	
2.1	FLAMMABLE GAS (LP gas, acetylene)		5.1	OXIDIZING AGENT (Calcium hypochlorite)	
2.2	NON-FLAMMABLE NON-TOXIC GAS (Carbon dioxide)		5.2	ORGANIC PEROXIDE	
2.3	TOXIC GAS (Chlorine gas)		6	TOXIC (Arsenic)	
3	FLAMMABLE LIQUID (Petrol, kerosene)		7	RADIOACTIVE MATERIAL (Uranium)	
4.1	FLAMMABLE SOLID (Firelighters, matches)		8	CORROSIVE (Hydrochloric acid)	
4.2	SPONTANEOUSLY COMBUSTIBLE (Carbon, white phosphorus)		9	MISCELLANEOUS DANGEROUS GOODS (Dry ice, asbestos)	
				MIXED CLASS LABEL (For road transport)	

FIGURE 20.2.6 HAZCHEM signs



FIGURE 20.2.7 Trucks transporting hazardous substances, such as flammable liquids, have hazard symbols attached.

Safety Data Sheets

Each chemical substance has an accompanying document called a **Safety Data Sheet (SDS)** (Figure 20.2.8), which was previously called a Material Safety Data Sheet (MSDS). An SDS contains important safety and first aid information about each chemical you commonly use in the laboratory. If the products of a reaction are toxic to the environment, you must pour your waste into a special container (not down the sink).

1. IDENTIFICATION OF THE MATERIAL AND SUPPLIER

Product Name: HYDROCHLORIC ACID - 20% OR GREATER

Recommended use of the chemical and restrictions on use: Precursor for generation of chlorine dioxide gas used in water treatment.

Supplier: Ixom Operations Pty Ltd
ABN: 51 600 546 512
Street Address: Level 8, 1 Nicholson Street
Melbourne 3000
Australia

Telephone Number: +61 3 9665 7111
Facsimile: +61 3 9665 7937
Emergency Telephone: 1 800 033 111 (ALL HOURS)

Please ensure you refer to the limitations of this Safety Data Sheet as set out in the "Other Information" section at the end of this Data Sheet.

2. HAZARDS IDENTIFICATION

Classified as Dangerous Goods by the criteria of the Australian Dangerous Goods Code (ADG Code) for Transport by Road and Rail; DANGEROUS GOODS.

This material is hazardous according to Safe Work Australia; HAZARDOUS SUBSTANCE.

Classification of the substance or mixture:

Corrosive to Metals - Category 1
Skin Corrosion - Sub-category 1B
Eye Damage - Category 1
Specific target organ toxicity (single exposure) - Category 3

SIGNAL WORD: DANGER



Hazard Statement(s):

H290 May be corrosive to metals.
H314 Causes severe skin burns and eye damage.
H335 May cause respiratory irritation.

Precautionary Statement(s):

Prevention:

P234 Keep only in original container.
P260 Do not breathe mist / vapours / spray.
P264 Wash hands thoroughly after handling.
P271 Use only outdoors or in a well-ventilated area.
P280 Wear protective gloves / protective clothing / eye protection / face protection.

FIGURE 20.2.8 Extracts of a Safety Data Sheet (SDS) for concentrated hydrochloric acid. The SDS alerts the reader to any potential hazards when using a substance, including appropriate measures to reduce risk of harm.

The SDS provides employers, workers and health and safety representatives with the necessary information to safely manage the risk of hazardous substance exposure.

Personal protective equipment

Everyone who works in a laboratory wears items that help keep them safe. This is called **personal protective equipment (PPE)** and includes:

- safety glasses
- shoes with covered tops
- disposable gloves when handling certain chemicals
- a disposable apron or a lab coat if there is risk of damage to clothing.

20.2 Review

SUMMARY

- The method of your investigation is a step-by-step procedure. When detailing the method, ensure it complies as a valid, reliable and accurate investigation.
- It is also important to determine how many times the experiment needs to be replicated. Many scientific investigations lack sufficient repetition to ensure that the results can be considered reliable and repeatable.
- It is important to choose appropriate equipment for your experiment. Select instrumentation that will give you accurate results.
- Risk assessments must be carried out before conducting an investigation to make sure that when you carry out your investigation you and others are kept safe. If you have elements of your investigation which are high risk you will need to revise your design.
- Appropriate personal protective equipment (PPE) that will help keep you safe should be used according to the risk assessment.

KEY QUESTIONS

- 1 A journal article reported the materials and method used in order to conduct an experiment. The experiment was repeated three times, and all values were reported in the results section of the article. Repeating an experiment and reporting results supports:
A validity
B reliability
C credibility
D systematic errors.
- 2 **a** Explain what is meant by the term 'control experiment'.
b Using an example, distinguish between independent and dependent variables.
- 3 You are conducting an experiment to determine the pH of various soft drinks. Identify:
a the independent variable
b the dependent variable
c at least one controlled variable.
- 4 You are conducting an experiment to determine the pH of a solution. Discuss the accuracy of your results if you are:
a using litmus paper or universal indicator
b recording the pH using a calibrated pH meter.

20.3 Uncertainty and error in data

During the planning and conducting of an experiment or investigation, minimising potential error is an important consideration.

IDENTIFYING ERRORS

Most practical experiments have errors associated with them. Errors can occur for a variety of reasons. Being aware of potential errors helps you to avoid or minimise them. For an investigation to be accurate, it is important to identify and record any errors.

There are two types of errors:

- systematic errors
- random errors.

When considering whether an error is random or systematic, you have to consider the measurement that is made compared to the true value of the value being measured.

Systematic errors

A **systematic error** will cause the measured error to be always above, or always below, the true value. It is an error that is consistent and will occur again if the investigation is repeated in the same way.

An example of a systematic error is if you were measuring the mass of a substance and on a piece of filter paper and you didn't take into account the mass of the filter paper. The mass recorded will always be higher than the true mass of just the substance being weighed. Another example would be if you were measuring the volume of a solution, and you transferred the solution from a beaker to a measuring cylinder to measure the volume and didn't ensure that all the solution was transferred. In this case the volume measured would always be less than the true volume of the solution in the beaker.

Systematic errors are usually a result of instruments that are not calibrated correctly or methods that are flawed.

Random errors

Random errors occur in an unpredictable manner and are generally small. They could give a value higher, or lower than the true value. Whenever a measuring device, whether it is a measuring cylinder or an electric balance is used, random error will occur.

Mistakes

Mistakes are not classified as errors. Mistakes could include misreading the numbers on a scale or spilling a portion of a sample. A measurement that involves a mistake must be rejected and not included in any calculations or averaged with other measurements of the same quantity.

TECHNIQUES FOR REDUCING ERROR

Designing the method carefully, including selection and use of equipment, will help reduce errors.

Appropriate equipment

Use the equipment that is best suited to the data to be collected to validate the hypothesis. Determining the units of the data being collected and at what scale will help to select the correct equipment. Using the right unit and scale will ensure that measurements are more accurate and precise (with smaller systematic errors).

Significant figures

Significant figures are the numbers that convey meaning and precision. The number of significant figures used depends on the scale of the instrument. It is important to record data to the number of significant figures available from the equipment or observation. Using either a greater or smaller number of significant figures can be misleading.

Review the following examples to learn more about significant figures:

- 15 has two significant figures
- 3.5 has two significant figures
- 3.50 has three significant figures
- 0.037 has two significant figures
- 1401 has four significant figures.

When adding or subtracting the sum or difference of a set of numbers should have as many numbers after the decimal as the least accurate in the set.

Adding: $5.0 + 3.21 + 1.2345 = 9.445 = 9.4$ (because 5.0 has only one figure after the decimal point)

Subtracting: $19.8765 - 1.23 = 18.6465 = 18.64$ (because 1.23.0 has only one figure after the decimal point)

When multiplying or dividing, the answer has no more significant figures than the number with the least significant figures.

Multiplying: $1.234 \times 2.34 = 2.88756 = 2.89$ (because 2.34 has only three significant figures)

Dividing: $34.56 \div 6.7 = 5.158208 = 5.2$ (because 6.7 has only two significant figures).

Calibrated equipment

Some equipment, such as pH meters, need to be calibrated before use. Before carrying out the investigation, make sure the instruments or measuring devices are properly calibrated and are, in general, functioning correctly. Record the precision of glassware that you intend to use. If you are preparing a solution of known concentration you might have access to a volumetric flask which has less uncertainty associated with measurements compared the using a beaker (Figure 20.3.1).



FIGURE 20.3.1 Record the uncertainty for glassware and instruments. This pipette can dispense a volume (aliquot) of 25.00 ± 0.03 mL. When used correctly, the volume dispensed will be between 24.97 mL and 25.03 mL.

Correct use of equipment

Use the equipment properly. Ensure any necessary training has been done to use the equipment and that you have had an opportunity to practice using the equipment before beginning the investigation. Improper use of equipment can result in inaccurate, imprecise data with large errors, and the validity of the data can be compromised.

Incorrect reading of measurements is a common misuse of equipment. Make sure all of the equipment required in the investigation can be used correctly and record the instructions in detail so they can be referred back to if the data doesn't appear correct.

Repeat trials

As discussed in Section 20.2, reliability is improved by repeating your experiment. Note that repeating trials will only reduce the effect of random errors. Repeating a flawed method will not reduce the effect of the systematic error within that method. Therefore modifications to your procedure may need to be looked at before repeating the investigation to ensure all variables are being tested under the same conditions.

20.3 Review

SUMMARY

- It is essential that during the investigation, the following are recorded in the logbook:
 - all quantitative and qualitative data collected
 - the methods used to collect the data
 - any incident, feature or unexpected event that may have affected the quality or validity of the data.
- A **systematic error** will cause the measured error to always above, or always below the true value.
- Random errors could give a value higher, or lower than the true value.
- The number of significant figures used depends on the scale of the instrument used. It is important to record data to the number of significant figures available from the equipment or observation.

KEY QUESTIONS

- 1 What type of error is associated with:
 - a inaccurate measurements?
 - b imprecise measurements?
- 2 Identify whether each of the situations below is a mistake, a systematic error or a random error.
 - a A student measured the mass of samples of octane in beakers and left the beakers uncovered in the lab before using them in an experiment.
 - b A student was measuring the mass of approximately 5 g samples of solid sodium chloride using a balance that was accurate to the nearest 1 g.
 - c A student sometimes forgot to re-zero the balance between weighing samples of solid sodium carbonate.
- 3 A scientist carries out a set of experiments, analyses the results and publishes them in a scientific journal. Other scientists in different laboratories repeat the experiment, but do not get the same results as the original scientist. Suggest three reasons that could explain this.

20.4 Processing data and information

Once you have conducted your investigation and collected data, you will need to find the best way of collating it. This section is a guide to the different forms of representation that will help you to better understand your data.

Raw data is unlikely to be used directly to validate the hypothesis. However, raw data is essential to the investigation and plans for collecting the raw data should be made carefully. Consider the formulas or graphs that will be used to analyse the data at the end of the investigation. This will help to determine the type of raw data that needs to be collected in order to validate the hypothesis.

Once you have determined the data that needs to be collected, prepare a table in which to record the data.

The raw data that has been obtained should be presented in a way that is clear, concise and accurate. There are a number of ways of presenting data, including tables, graphs, flow charts and diagrams.

PRESENTING RAW AND PROCESSED DATA IN TABLES

Tables organise data into rows and columns and can vary in complexity according to the nature of the data. Tables can be used to organise raw data and processed data or to summarise results.

The simplest form of a table is a two-column format. In a two-column table, the first column should contain the independent variable (the one being changed) and the second column should contain the dependent variable (the one that may change in response to a change in the independent variable).

Tables should have the following features:

- a descriptive title
- column headings (including the unit)
- the independent variable placed in the left column
- the dependent variable placed in the right column.

Look at Table 20.4.1, which has been used to organise raw and processed data about the water near the desalination plant in Cockburn Sound.

TABLE 20.4.1 Analysis of seawater from Cockburn Sound and how the data could be recorded

Location	Temperature (°C)	pH	Conductivity (mS m ⁻¹)	Turbidity (NTU)	Dissolved oxygen (DO) (mg L ⁻¹)	Biological oxygen demand (BOD) (mg L ⁻¹)
A	10.4	6.5	72	5	11.8	0.9
B	9.2	6.6	73	10	11.4	0.7
C	9.5	6.4	77	10	10.9	0.9
D	9.9	6.5	75	10	11.3	1.0

A table of processed data usually presents the average values of replicates: the **mean**.

GRAPHS

In general, tables provide more detailed data than graphs, but it is easier to observe trends and patterns in data in graphical form than in tabular form.

Graphs are used when two variables are being considered and one variable is dependent on the other. The graph shows the relationship between the variables.

There are several types of graphs that can be used, including line graphs, bar graphs and pie charts. The best one to use will depend on the nature of the data.

General rules to follow when making a graph (Figure 20.4.1) include the following.

- Keep the graph simple and uncluttered.
- Use a descriptive title.
- Represent the independent variable on the x -axis and the dependent variable on the y -axis.
- Make axes proportionate to the data.
- Clearly label axes with both the variable and the unit in which it is measured.

Line graphs

Line graphs are a good way of representing continuous quantitative data. In a line graph, the values are plotted as a series of points on the graph. There are two ways of joining these points.

- A line can be ruled from each point to the next. It shows the overall trend; it is not meant to predict the value of the points between the plotted data.
- The points can be joined with a single smooth straight or curved line (Figure 20.4.2). This creates a **trend line**, also known as a line of best fit. The line of best fit does not have to pass through every point but should go close to as many points as possible. It is used when there is an obvious trend between the variables.

Sometimes when the data is collected, there may be one point that does not fit with the trend and is clearly an error. This is called an outlier. An outlier is often caused by a mistake made in measuring or recording data. If there is an outlier, include it on the graph, but ignore it when adding a line of best fit. As you can see in Figure 20.4.2, the point (0.48, 0.08) is an outlier because it does not fit the trend of the other data points therefore it was not used in drawing the line of best fit.

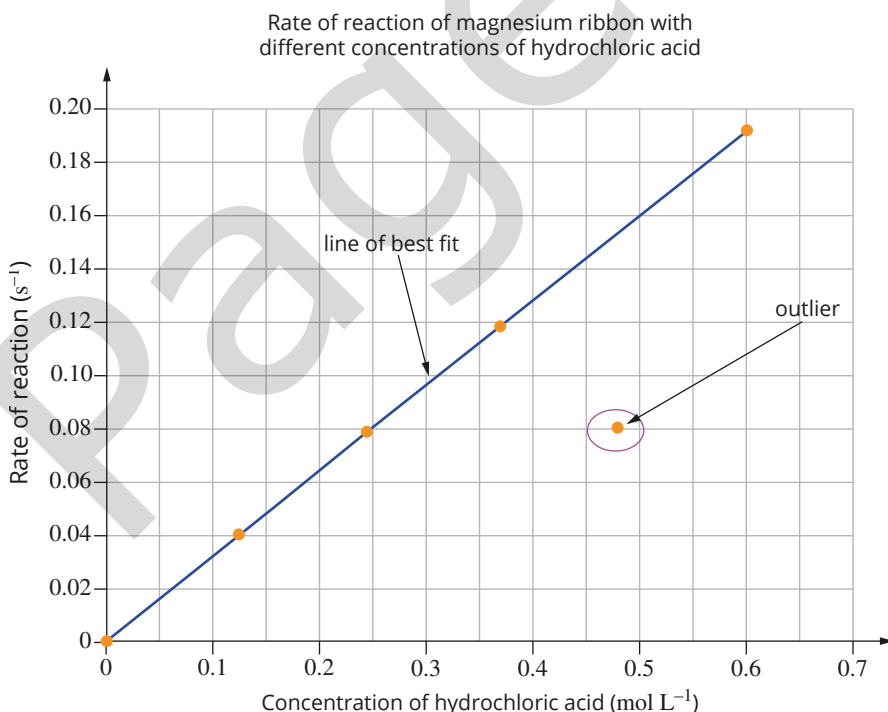


FIGURE 20.4.2 The graph shows an example of an outlier ignored in forming a line of best fit.

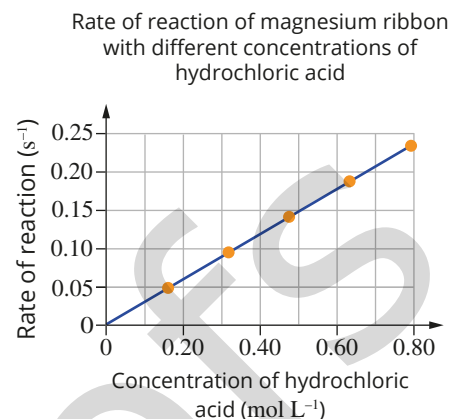


FIGURE 20.4.1 A graph is a better way to observe trends and patterns in the data.

20.4 Review

SUMMARY

- Consider how the data will be presented and analysed. A wide range of analysis tools could be used. For example, tables organise data so that patterns can be established and graphs can show relationships and comparisons.
- The simplest form of a table is a two-column format in which the first column contains the independent variable and the second column contains the dependent variable
- General rules to follow when making a graph include the following.
 - Keep the graph simple and uncluttered.
 - Use a descriptive title.
 - Represent the independent variable on the x-axis (horizontal) and the dependent variable on the y-axis (vertical).
 - Make axes proportionate to the data.
 - Clearly label axes with both the variable and the unit in which it is measured.

KEY QUESTIONS

- Determine the mean for the data set: {21, 28, 19, 19, 25, 24}.
- How can the general pattern (trend) of a graph be represented once the points are plotted?
- Use the following values to plot a graph showing the rate of reaction between sodium hydrogencarbonate and solutions of citric acid with a range of concentrations. The rate of reaction was measured by recording the volume of carbon dioxide produced in the first 3 seconds of the experiment.

Concentration of citric acid (mol L^{-1})	Volume of CO_2 produced (mL)
0.10	5
0.20	10
0.30	14.5
0.40	20.5
0.50	21.5
0.60	30

- From the graph you have drawn, select the data point that is an outlier.
- Define the term 'outlier'.

20.5 Analysing data and information

Now that the investigation has been conducted and data collected, it is time to draw it all together. You will now analyse your results to better understand the chemical processes behind them.



FIGURE 20.5.1 To discuss and conclude your investigation, utilise the raw and processed data.

ANALYSING AND EVALUATING DATA

In the discussion, the findings of the investigation need to be analysed and interpreted.

- State whether a pattern, trend or relationship was observed between the independent and dependent variables. Describe what kind of pattern it was and specify under what conditions it was observed.
- Were there discrepancies, deviations or anomalies in the data? If so, these should be acknowledged and explained.
- Identify any limitations in the data you have collected. Perhaps a larger sample or further variations in the independent variable would lead to a stronger conclusion.

Trends in line graphs

Graphs are drawn to show the relationship, or trend, between two variables.

- Variables that change in linear or direct proportion to each other produce a straight, sloping trend line (Figure 20.5.2).
- Variables that change exponentially in proportion to each other produce a curved trend line (Figure 20.5.3).
- When there is an inverse relationship, one variable increases as the other variable decreases (Figure 20.5.4).
- When there is no relationship between two variables, one variable will not change even if the other changes (Figure 20.5.5).

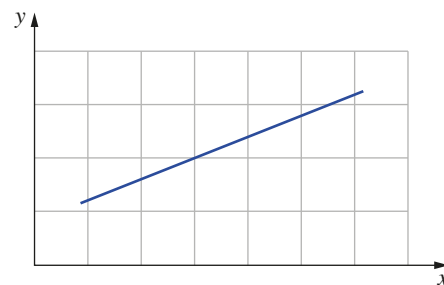


FIGURE 20.5.2 Trend line for variables that change in direct response to each other

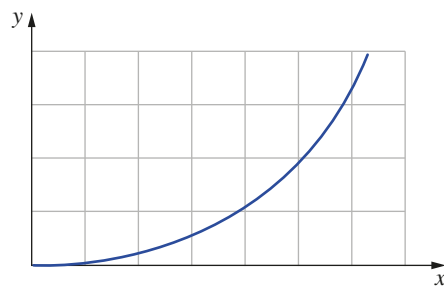


FIGURE 20.5.3 Variables that change in response to each other in a non-linear way

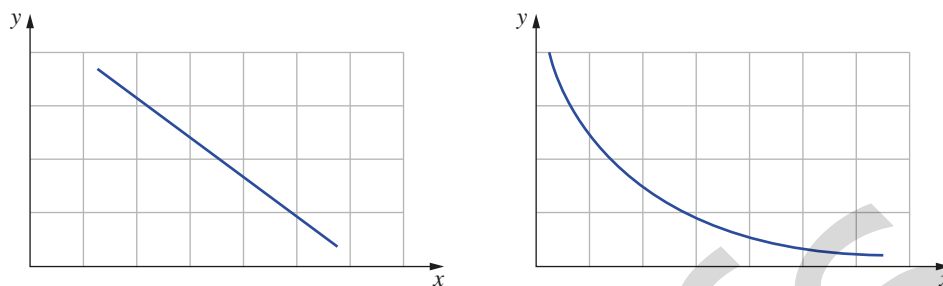


FIGURE 20.5.4 An inverse relationship in which one variable decreases in response to the other variable increasing. It may be (a) direct or (b) non-linear.

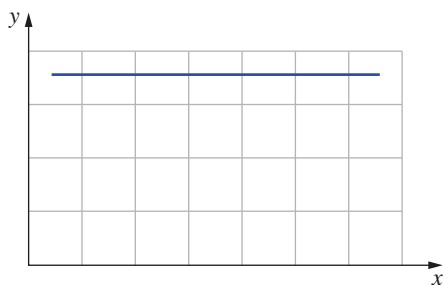


FIGURE 20.5.5 When two variables show no relationship, there is no trend in the data.

Remember that the results might be unexpected. This does not make the investigation a failure. However, the findings must be related to the hypothesis, aims and method.

Table 20.5.1 lists the types of graphs used for various examples.

TABLE 20.5.1 Examples of the types of graphs that could be used in your report

Type of graph	When to use	Example
scatter graph	when showing quantitative data where one variable is dependent on another variable; draw a line of best fit to show the relationship between the two variables	showing the relationship between the rate of a reaction and the concentration of acid.
line graph	with continuous quantitative data	concentration of dissolved oxygen at a particular location of a creek over a period of time
bar graphs	when comparing data in an investigation with a qualitative independent variable	turbidity of water at various locations, as shown in Figure 20.5.6
pie diagrams	when summarising qualitative data; to display proportions	relative proportion of different pesticides in a sample of water

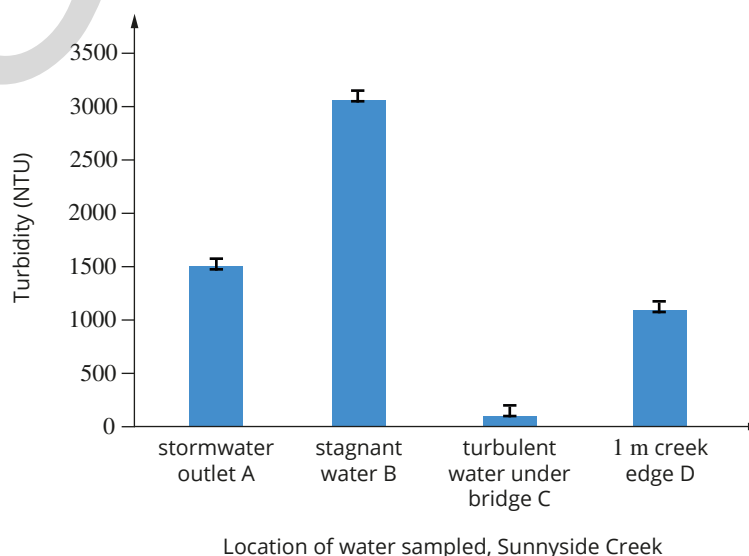


FIGURE 20.5.6 A bar graph can be used to compare data. This graph shows a measure of water quality, called turbidity, of water samples taken from various locations along Sunnyside Creek.

EVALUATING THE METHOD

It is important to discuss the limitations of the investigation method. Evaluate the method and identify any issues that could have affected the validity, accuracy, precision or reliability of the data. Sources of errors and uncertainty must also be stated in the discussion.

Once any limitations or problems in the methodology have been identified, recommend improvements on how the investigation could be conducted if repeated; for example, suggest how a journal article reported the materials could be minimised or eliminated.

Bias

Bias may occur in any part of the investigation method, including sampling and measurements.

Bias is a form of systematic error resulting from the researcher's personal preferences or motivations. There are many types of bias, including:

- poor definitions of both concepts and variables (for example, not defining pH)
- incorrect assumptions (for example, measuring pH of a solution without also measuring the temperature of the solution)
- errors in the investigation design and methodology (for example, taking water samples from more than one site).

Some biases cannot be eliminated, but should be addressed in the discussion.

Accuracy and precision

In the discussion, evaluate the degree of accuracy and precision of the measurements for each variable of the hypothesis. Comment on the uncertainties obtained.

When relevant, compare the chosen method with any other methods that might have been selected, evaluating the advantages and disadvantages of the selected method and the effect on the results.

Reliability

When discussing the results, indicate the range of the data obtained from replicates. Explain how the sample size was selected. Larger samples are usually more reliable, but short timeframes and scarce resources might have prevented multiple sets of data from being collected. Discuss whether the results of the investigation have been limited by the sample size.

The control group is important to the reliability of the investigation. A control group helps determine if a variable that should have been controlled has been overlooked and might explain any unexpected results.

Error

Discuss any source of systematic or random error and suggest ways of improving the investigation.

CRITICALLY EVALUATING SOURCES OF INFORMATION

Not all sources are **credible**. When you have used primary or secondary data from information sources it is essential to critically evaluate the content and its origin. Questions you should always ask when evaluating a source include the following.

- Who created this source? What are the qualifications, **expertise**, **reputation** and **affiliation** of the authors?
- Why was it written?
- Where was the information published?
- When was the information published?
- Are conclusions supported by data or evidence?
- What is implied?
- What is omitted?

- Are any opinions or values being presented in the piece?
- Is the writing objectively and accurately describing a scientific concept or **phenomenon**?
- How might other people understand or interpret this message differently from me?
- When evaluating the validity or bias of websites consider its domain extension:
 - .gov stands for government
 - .edu stands for education
 - .org stands for non-profit organisations
 - .com stands for commercial/business.



FIGURE 20.5.7 Honest evaluation and reflection play important roles in analysing methodology.

20.5 Review

SUMMARY

- After completing your investigation you will need to analyse and interpret your data. A discussion of your results is required where the findings of the investigation need to be analysed and interpreted.
- State whether a pattern, trend or relationship was observed between the independent and dependent variables. Describe what kind of pattern it was and specify under what conditions it was observed.
- Were there discrepancies, deviations or anomalies in the data? If so, these should be acknowledged and explained.
- Identify any limitations in the data collected. Perhaps a larger sample or further variations in the independent variable would lead to a stronger conclusion.
- It is important to discuss the limitations of the investigation method. Evaluate the method and identify any issues that could have affected the validity, accuracy, precision or reliability of the data. Sources of errors and uncertainty must also be stated in the discussion.
- When discussing the results, indicate the range of the data obtained from replicates. Explain how the sample size was selected. Larger samples are usually more reliable, but time and resources might have been scarce. Discuss whether the results of the investigation have been limited by the sample size.

KEY QUESTIONS

- 1 What relationship between the variables is indicated by a sloping linear graph?
- 2 What relationship exists if one variable decreases as the other increases?
- 3 What relationship exists if both variables increase or both decrease at the same rate?
- 4 What might cause a sample size to be limited in an investigation?

20.6 Conclusions

Having analysed your results you can then apply them to chemical concepts in order to evaluate your conclusions. In this section you will learn how analysing your investigation leads to a better understanding of the underlying scientific principles of your research.

DISCUSSING RELEVANT CHEMICAL CONCEPTS

To make the investigation more meaningful, it should be explained within the right context, meaning the related chemical ideas, concepts, theories and models. Within this context, explain the basis for the hypothesis. For example, if you were studying the impact of dissolved carbon dioxide on the pH of sparkling mineral water you could include the information in Table 20.6.1 in your discussion.

TABLE 20.6.1 Examples of how to include chemical concepts in your discussion

Key ideas	Example
definitions of key terms	'pH', 'dissolved carbon dioxide' and 'sparkling' mineral water
the function of added carbon dioxide	in order to create 'sparkling' water
relationship between variables	dissolved carbon dioxide and pH of water, temperature was controlled in the experiment
chemical principles	dissolved carbon dioxide and formation of carbonic acid (H_2CO_3), including relevant equations
sources of error	reducing random error, by repeating measurements and calculating average

Relating your findings to a chemical concept

In Section 20.5 you learnt that during the analysing stage of your investigation, you should look for trends, patterns and mathematical models in your results. This is the framework needed to discuss whether the data supported or refuted the hypothesis. You should ask questions such as the following.

- Was the hypothesis supported?
- Has the hypothesis been fully answered? (If not, give an explanation of why this is so and suggest what could be done to either improve or complement the investigation.)
- Do the results contradict the hypothesis? If so, why? (The explanation must be plausible and must be based on the results and previous evidence.)

Be sure to discuss the broader implications of the findings. Implications are the bigger picture. Outlining them for the audience is an important part of the investigation. Ask questions like these.

- Do the findings lead to further questions?
- Can the findings be extended to another situation?
- Are there any practical applications for the findings?

DRAWING EVIDENCE-BASED CONCLUSIONS

A conclusion is usually written as a paragraph that links the collected evidence to the hypothesis and provides a justified response to the research question.

Indicate whether the hypothesis was supported or refuted and the evidence on which this decision is based (that is, the results). Do not provide irrelevant information. Only refer to the specifics of the hypothesis and the research question and do not make generalisations.

Examples of poor and better conclusions for the hypothesis and research question given are listed in Table 20.6.2 and Table 20.6.3.

TABLE 20.6.2 Examples of strong and weak conclusions to a hypothesis

Hypothesis: A decrease in the concentration of solution of hydrochloric acid will result in an increase in the measured pH of the solution.

Strong conclusion	Weak conclusion
A decrease in the concentration of the acid by a factor of 10 resulted in a decrease in the pH of the acid solution from 1.5 to 2.5.	The pH of the solution increased as the hydrochloric acid concentration decreased.

TABLE 20.6.3 Examples of strong and weak conclusions in response to a research question

Research question: How is the rate of the reaction between magnesium ribbon and an acid related to the temperature of the acid?

Strong conclusion	Weak conclusion
Analysis of the results on the effect of increasing the temperature of the acid suggest that each increase of 10°C results in an approximate doubling of the rate of the reaction. This supports the collision theory, which states that higher temperature increases both the number of collisions and the proportion of successful collisions.	The results show that the reaction is faster at higher temperatures.

20.6 Review

SUMMARY

- To make the investigation more meaningful, it should be explained within the right context, meaning the related chemical ideas, concepts, theories and models. Within this context, explain the basis for the hypothesis.
- Indicate whether the hypothesis was supported or refuted and on what evidence this decision is based (that is, the results). Do not provide irrelevant information. Only refer to the specifics of the hypothesis and the research question and do not make generalisations.

KEY QUESTIONS

- Which of the following would **not** support a strong conclusion to a report?
 - The concluding paragraphs are relevant and provide evidence.
 - The concluding paragraphs are written in emotive language.
 - The concluding paragraphs include reference to limitations of the research.
 - The concluding paragraphs include suggestions for further avenues of research.
- A procedure was repeated 30 times. How should the following statement be rewritten?

Many repeats of the procedure were conducted.

20.7 Communicating

How you approach communicating your results will depend on the audience you want to reach. If you are communicating with a general audience you may want to discuss your investigation in the style of a news article or blog post. These types of communication use simple language and analogies, and don't use too much scientific language as you need to assume that your audience does not have a science background.

Throughout this course you will need to present your research using appropriate scientific language and notation. There are many different presentation formats that you can use, such as posters, oral presentations and reports. This section covers the main characteristics of effective science communication and report writing, including objectivity, clarity, conciseness and coherence.

STRUCTURING A REPORT

Your report should have a clear, logical structure.

Introduction:

- The first paragraph should introduce your topic and define key terms.

Body paragraphs:

- Each subsequent paragraph should cover one main idea.
- Use evidence to support statements.
- Avoid very long or very short paragraphs.

Conclusion:

- The final section should summarise the main findings.
- It should relate to the aim of the investigation.
- The conclusion should include limitations and should discuss implications and applications of the research and potential future research.

Analysing information relevant to your research investigation

Scientific research should always be objective and neutral. Any premise presented must be supported with facts and evidence to allow the audience to make its own decision. Identify the evidence supporting or contradicting each point you want to make. It is also important to explain connections between ideas, concepts, theories and models. Figure 20.7.1 lists the questions you need to consider when writing your investigation report.

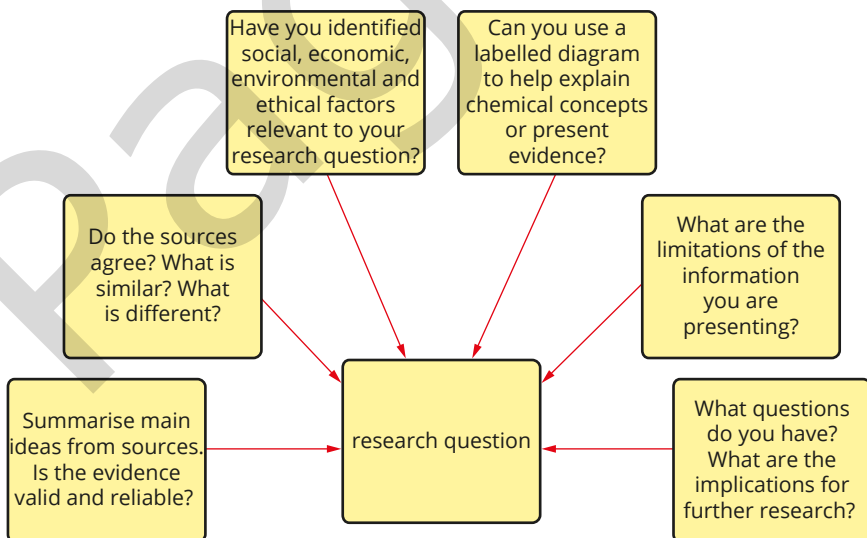


FIGURE 20.7.1 Discuss relevant information, ideas, concepts and implications, and make sure your discussion is relevant to the question under investigation.

WRITING FOR SCIENCE

Scientific reports are usually written in an objective or unbiased style. This is in contrast with English writing that most often uses the subjective techniques of rhetoric or persuasion. Table 20.7.1 contrasts persuasive and scientific writing styles.

TABLE 20.7.1 Persuasive writing versus scientific writing styles

Persuasive writing examples	Scientific writing equivalent examples
Use of biased and subjective language Examples: The results were extremely bad, atrocious, wonderful etc. This is terrible because ... This produced a disgusting odour. Health crisis	Use of unbiased and objective language Examples: The results showed ... The implications of these results suggest ... The results imply ... This produced a foul odour. Health issue
Use of exaggeration Example: The object weighed a colossal amount, like an elephant.	Use of non-emotive language Example: The object weighed 256 kg.
Use of everyday or colloquial language Examples: The bacteria passed away. The results don't ... The researchers had a sneaking suspicion ...	Use of formal language Examples: The bacteria died. The results do not ... The researchers predicted/hypothesised/theorised ...

Consistent reporting narrative

Scientific writing can be written either in first-person or in third-person narrative. Your teacher may advise you on which to select. In either case, ensure that you keep the narrative point of view consistent. Table 20.7.2 gives examples of first-person and third-person narrative.

TABLE 20.7.2 Examples of first-person and third-person narrative

First person	Third person
I put 50.0g marble chips in a conical flask and then added 10.0 mL of 2 mol L ⁻¹ hydrochloric acid.	First, 50.0g of marble chips was weighed into the conical flask and then 10.0 mL of 2.0 mol L ⁻¹ hydrochloric acid was added.
After I observed the reaction, I found that ...	After the reaction was completed, the results showed ...
My colleagues and I found ...	Researchers found ...

Qualified writing

Be careful of words that are absolute, such as *always*, *never*, *shall*, *will* and *proven*. Sometimes it may be more accurate and appropriate to use qualifying words, such as *may*, *might*, *possible*, *probably*, *likely*, *suggests*, *indicates*, *appears*, *tends*, *can* and *could*.

Concise writing

It is important to write concisely, particularly if you want to engage and maintain the interest of your audience. Use shorter sentences. Table 20.7.3 shows some examples of more concise wording.

TABLE 20.7.3 Examples of verbose and concise language

Verbose example	Concise example
Due to the fact that	Because
Smith undertook an investigation into ...	Smith investigated
It is possible that the cause could be ...	The cause might be ...
A total of five experiments	Five experiments
End result	Result
In the event that	If
Shorter in length	Shorter

Visual support

Identify concepts that can be explained using visual models and information that can be presented in graphs or diagrams. This will not only reduce the word count of your work but will also make it more accessible for your audience.

For example, flow charts convey the steps in a process or method. The flow chart in Figure 20.7.2 shows how polymers used in the production of a consumer item can be decomposed. A limitation of flow charts is that the details of the process are omitted. Of course, simplification is often a benefit of visual models.

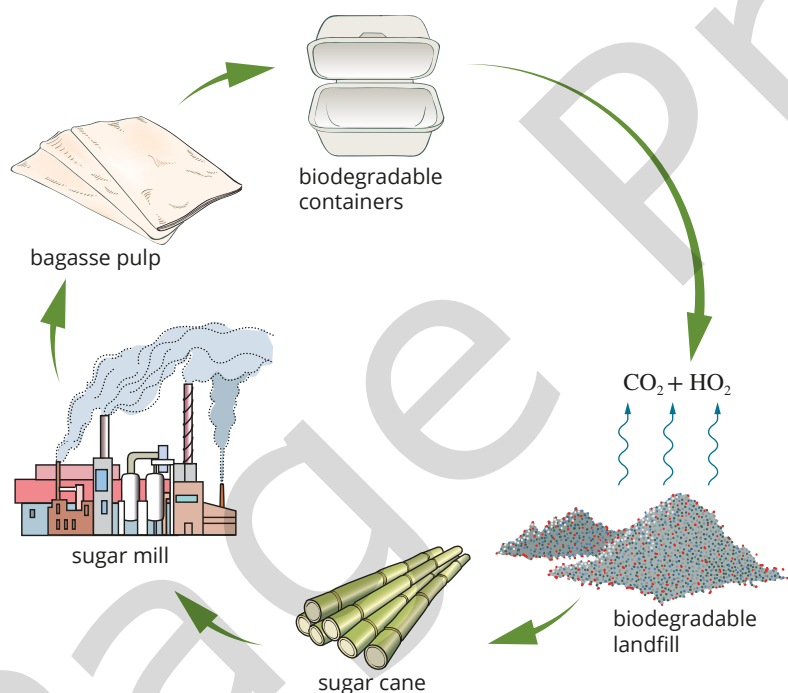


FIGURE 20.7.2 This flow chart shows how polymers used in the production of a consumer item can be decomposed. A limitation of this diagram is that it does not indicate the timeline or details involved in this process.

EDITING YOUR REPORT

Editing your report is an important part of the process. After editing your report, save new drafts with a different file name and always back up your files in another location.

Pretend you are reading your report for the first time when editing. Once you have completed a draft, it is a good idea to put it aside and return to it with ‘fresh eyes’ a day later. This will help you find areas that need further work and give you the opportunity to improve them. Look for content that is:

- ambiguous or unclear
- repetitive

- awkwardly phrased
- too lengthy
- not relevant to your research question
- poorly structured
- lacking evidence
- lacking a reference (if it is another researcher's work).

Use a spellchecker tool to help you identify typographical errors, but first, check that your spellchecker is set to Australian English.

REFERENCES AND ACKNOWLEDGEMENTS

All the quotations, documents, publications and ideas used in the investigation need to be acknowledged in the 'references and acknowledgments' section in order to avoid plagiarism and to ensure authors are credited for their work. References and acknowledgements also give credibility to the study and allow the audience to locate information sources should they wish to study it further. The standard referencing style used is the American Psychological Association (APA) academic referencing style.

When referencing a book, include in this order:

- author's surname and initials
- date of publication
- title
- publisher's name
- place of publication.

For example: Rickard, Greg, et al. (2016), *Pearson Science SB9* (2nd ed.), Pearson Education, Melbourne, Australia.

When referencing a website, include in this order:

- author's surname and initials, or name of organisation, or title
- year website was written or last revised
- title of webpage
- date website was accessed
- website address.

For example: National Geographic (2015), 'Killer fungus that's devastating bats may have met its match', accessed 29 May 2015, from <http://news.nationalgeographic.com/2015/05/150527-bats-white-nose-syndrome-treatment-conservation-animals-science>

In-text citations

Each time you write about the findings of other people or organisations, you need to provide an in-text citation and provide full details of the source in a reference list. In the APA style, in-text citations include the first author's last name and date in brackets (author, date). List the full details in your list of references.

The following examples show the use of in-text citation.

It was reported that in testing of five pro-oxidant additives added to commonly manufactured polymers, none resulted in significant biodegradation after three years (Selke et al., 2015).

Or

Selke et al. (2015) reported that in testing of five pro-oxidant additives added to commonly manufactured polymers, none resulted in significant biodegradation after three years.

The bibliographic details of the example above would be:

Selke, S., Auras, R., Nguyen, T.A., Aguirre, E.C., Cheruvathur, R., & Liu, Y. (2015). Evaluation of biodegradation—promoting additives for plastics. *Environmental Science & Technology*, 49(6), 3769–3777.

CHEMICAL NOMENCLATURE

Using appropriate chemical nomenclature, scientific notations and units is also important. Table 20.7.4 has examples of the most common terms used in chemistry.

TABLE 20.7.4 The most common terms used in chemistry

Chemical term	Definition	Examples
element	substance whose atoms have the same atomic number; atoms of different elements have different atomic numbers	sodium, chlorine, tin
compound	substance consisting of two or more elements combined in fixed proportions; a chemical formula can be written for a pure compound	H ₂ O, NaCl
substance	element, compound	H ₂ , diamond, H ₂ O
particle	an atom, ion, molecule, proton, neutron, electron	Ne, Na ⁺ , H ₂ O, H ⁺ , ¹² C, e ⁻
atom	building block of matter; the smallest unit of an element	Na, He, C, Sn
molecule	two or more atoms covalently bonded together	H ₂ O, C ₄ H ₁₀ , C ₆ H ₁₂ O ₆ water, butane, glucose
ion	a positively charged or negatively charged atom or group of atoms resulting from the loss or gain of one or more electrons.	Na ⁺ , Cl ⁻ , NO ₃ ⁻
cation	positively charged ion	Na ⁺ , Mg ²⁺ , Al ³⁺ , NH ₄ ⁺
anion	negatively charged ion	Cl ⁻ , O ²⁻ , PO ₄ ³⁻
network layer lattice	covalently bonded layer lattice	graphite
network lattice	covalently bonded lattice in which the bonds extend in three dimensions	diamond, silicon dioxide

Table 20.7.5 describes some chemical nomenclature (naming conventions) commonly used in chemistry.

TABLE 20.7.5 Chemical nomenclature

Application	Convention	Examples
naming elements	the first letter is not capitalised unless at the start of a sentence	carbon, hydrogen
element symbols	first letter is capitalised, subsequent letter (if present) is lower case	N, Na, Ne, Ni
naming metals that can form ions of different charges (oxidation states)	write the name of the metal, followed by a Roman numeral in brackets	Cu ⁺ , name is copper(I) ion Cu ²⁺ , name is copper(II) ion
naming ionic compounds	name the cation before the anion	sodium chloride tin(IV) chloride
writing ionic formulae	write the cation before the anion	NaCl SnCl ₄
use of brackets in chemical formulae	to indicate atoms that need to be considered together	Al ₂ (CO ₃) ₃
use of brackets in condensed structural (semistructural) formulae	can indicate which groups of atoms: • are attached to a central atom (such as a carbon atom); • is repeated, for example repeating CH ₂ groups	CH ₃ CH(CH ₃)CH ₃ CH ₃ (CH ₂) ₅ CH ₃

Examples of uses and limitations of some commonly used chemical conventions and representations are presented in Table 20.7.6.

TABLE 20.7.6 Examples of uses and limitations of some commonly used chemical conventions and representations

Representation	Use	Limitations	Example
molecular formula	gives the actual number of atoms of each element in a molecule of a compound	does not show the arrangement of atoms in a molecule	C_4H_{10} , H_2O
structural formula	shows the relative location of atoms within a molecule in two dimensions	does not show the arrangement of atoms in three dimensions	propene structural formula <i>methanol</i> <i>chloroethane</i> <i>propane</i> <i>ethanoic acid</i> <i>propene</i>
condensed structural (semistructural) formula	enables a structural formula to be drawn	does not show the arrangement of atoms in three dimensions	$CH_3CH_2CH_2CH_3$ or $CH_3(CH_2)_2CH_3$

MEASUREMENT AND UNITS

In every area of chemistry we have attempted to quantify the phenomena we study. In investigations we generally make measurements and process those measurements in order to come to some conclusions. Scientists have a number of conventional ways of interpreting and analysing data from their investigations. There are also conventional ways of writing numerical measurements and their units.

Table 20.7.7 presents the main quantities and units used in chemistry.

TABLE 20.7.7 Quantities and units used in chemistry

Quantity	Symbol for quantity	Unit
Mass	m	gram (g)
Volume	V	litre (L)
Amount of substance	n	mole (mol)
Molar mass	M	grams per mole ($g\ mol^{-1}$)
Relative molecular or formula mass	M_r	no units, relative to one atom of ^{12}C exactly
Relative atomic mass	A_r	no units, relative to one atom of ^{12}C exactly
Density	d	$g\ mL^{-1}$, $kg\ L^{-1}$
Molarity	c	moles per litre ($mol\ L^{-1}$)

Correct use of unit symbols

The correct use of unit symbols removes ambiguity, as symbols are recognised internationally. The symbols for units are not abbreviations and should not be followed by a full stop unless they are at the end of a sentence.

Upper-case letters are only used for the *symbols* of the units that are named after people. For example, the unit of energy is joule and the symbol is J. The joule was named after James Joule who was famous for studies into energy conversions. The exception to this rule is ‘L’ for litre. We do this because a lower-case ‘l’ looks like the numeral ‘1’.

The product of a number of units is shown by separating the symbol for each unit with a space. The division or ratio of two or more units can be shown in fraction form, using a slash, or using negative indices. Prefixes should not be separated by a space.

Table 20.7.8 provides some examples of correct and incorrect use of symbols for derived units.

TABLE 20.7.8 Some examples of the use of symbols for derived units

Preferred	Incorrect
g mol ⁻¹	gmol ⁻¹
kPa	KPa

Units named after people can take the plural form by adding an 's' when used with numbers greater than one. *Never* do this with the unit symbols. It is acceptable to say 'two kilojoules' but it is wrong to write 2 kJs.

Numbers and symbols should not be mixed with words for units and numbers. For example, thirty grams and 30g are correct while 30 grams and thirty g are incorrect.

Scientific notation

To overcome confusion or ambiguity, measurements are often written in scientific notation. Quantities are written as a number between one and ten and then multiplied by an appropriate power of ten. Note that ‘scientific notation’, ‘standard notation’ and ‘standard form’ all have the same meaning.

Examples of some values written in scientific notation are:

$$6.022 \times 10^{23} \text{ particles}$$

$$25.25 \text{ mL} = 2.525 \times 10^{-2} \text{ L}$$

$$0.00302 \text{ mol} = 3.02 \times 10^{-3} \text{ mol}$$

You should be routinely using scientific notation to express numbers. This also involves learning to use your calculator intelligently. Scientific and graphics calculators (Figure 20.7.3) can be put into a mode whereby all numbers are displayed in scientific notation. It is useful when doing calculations to use this mode rather than frequently attempting to convert to scientific notation by counting digits on the calculator display.

An important reason for using scientific notation is that it removes ambiguity about the precision of some measurements. For example, a measurement recorded as 240 g could be a measurement to the nearest gram; that is, somewhere between 239.5 g and 240.5 g, or it could have been rounded up to the nearest 10 g. By writing this quantity as 2.40×10^2 g it makes it clear the accuracy of the value (in this case, 3 significant figures).

Prefixes and conversion factors

Conversion factors should be used carefully. You should be familiar with the prefixes and conversion factors in Table 20.7.9. Note that the table gives all conversions as a multiplying factor.



FIGURE 20.7.3 A scientific calculator

TABLE 20.7.9 Prefixes and conversion factors

Multiplying factor		Prefix	Symbol
1 000 000 000 000	10^{12}	tera	T
1 000 000 000	10^9	giga	G
1 000 000	10^6	mega	M
1 000	10^3	kilo	k
0.01	10^{-2}	centi	c
0.001	10^{-3}	milli	m
0.000 001	10^{-6}	micro	μ
0.000 000 001	10^{-9}	nano	n
0.000 000 000 001	10^{-12}	pico	p

Do not put spaces between prefixes and unit symbols.

It is important to give the symbol the correct case (upper or lower case). There is a big difference between 1 mm and 1 Mm.

20.7 Review

SUMMARY

- A scientific report must include an introduction, body paragraphs and conclusion.
- The conclusion should include a summary of the main findings, a conclusion related to the issue being investigated, limitations of the research, implications and applications of the research, and potential future research.
- Scientific writing uses unbiased, objective, accurate, formal language. Scientific writing should also be concise and qualified.
- Visual support can assist in conveying scientific concepts and processes efficiently.
- Ensure you edit your final report.
- Scientific notation needs to be used when communicating your results.

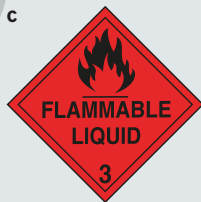
KEY QUESTIONS

- Which of the following statements is written in scientific style?
 - The results were fantastic ...
 - The data in Table 2 indicates ...
 - The researchers felt ...
 - The smell was awful ...
- Which of the following statements is written in first-person narrative?
 - The researchers reported ...
 - Samples were analysed using ...
 - The experiment was repeated three times ...
 - I reported ...
- The variables molar mass and molarity each have different units. Write the units for each of the following in correct notation.
 - molar mass; grams per mole
 - molarity; moles per litre
- Show how to convert grams into kilograms.
- Discuss why you might need to convert between different multiplying factors, for example litres to millilitres.

Chapter review

KEY TERMS

accuracy	independent variable	qualitative	
bias	mean	quantitative	
chemical (HAZCHEM) codes	outlier	random error	
control experiment	personal protective	raw data	systematic error
controlled variable	equipment (PPE)	reliability	trend line
credible	phenomenon	Safety Data Sheet (SDS)	uncertainty
dependent variable	precision	secondary source	validity
hypothesis	primary source	significant figures	variable

- Consider the following research question:
'Is the concentration of lead in water sampled from Esperance Harbour within acceptable limits?'
Which of the following are the independent, dependent and controlled variables?
 - concentration of lead
 - analytical technique, temperature of water sample, type of sampling container
 - source and location of water
- Match the following command verbs with their definitions: describe, analyse, apply, create, identify, reflect, investigate.
 - think deeply about
 - produce or make new
 - identify connections and relationships, interpret to reach a conclusion
 - observe, study, examine, inquire systematically in order to establish facts or derive conclusions
 - use knowledge and understanding in a new situation
 - recognise or indicate what or who
 - give a detailed account.
- Consider the following hypothesis:
'The concentration of heavy metals in industrial waste water will be greater than that of drinking water.'
Name the independent, dependent and controlled variables for an experiment with this hypothesis.
- Which graph from the following list would be best to use with each set of data listed here?
Graph types: pie diagram, scatter graph (with line of best fit), bar graph, line graph
 - the levels of a pesticide detected in drinking water at various locations
 - the temperature of water sampled at the same time of day over a period of a month
 - a calibration curve showing conductivity of a solution compared to the concentration of dissolved ions
 - the proportion of specific contaminants detected in water.
- What is the meaning of each of the following chemical codes?
 - 
 - 
 - 
- Explain the terms 'accuracy' and 'validity'.
- Identify whether the following are mistakes, systematic errors or random errors.
 - A student spills some solution during an experiment.
 - The reported measurements are above and below the true value.
 - A weighing balance has not been calibrated.
- What relationship between variables is indicated by a curved trend line?
- A scientist designed and completed an experiment to test the following hypothesis: 'Increasing the temperature of water would result in an increase in the measured electrical conductivity of water.'
The discussion section of the scientist's report included comments on the reliability, validity, accuracy and precision of the investigation.
Determine if the following sentences comment on the reliability, validity, accuracy or precision.
 - Three water samples from the same source were examined at each temperature. Each water sample was analysed and the measurements were recorded.
 - The temperature and the electrical conductivity of the water samples were recorded using data-logging equipment. The temperature of some of the water samples was measured using a glass thermometer.
 - The data-logging equipment was calibrated for electrical conductivity against a known standard. The equipment was calibrated before measurements were taken.
 - The temperature probe (data logger) measured temperature to the nearest 0.1°C. The glass thermometer measured temperature to the nearest 1°C.

- 10** What factors should you consider to ensure you discuss the limitations of your method?
- 11** Explain the meaning of the term ‘trend’ in a scientific investigation and describe the types of trends that might exist.
- 12** What is meant by the ‘limitations’ of the investigation method?
- 13** What is ‘bias’ in an investigation?
- 14** Which of the following consists only of secondary sources of information?
- A** An interactive periodic table, an article published in a science magazine, a science documentary, a practical report written by a Year 11 student
 - B** An article published in a peer-reviewed science journal, an article published in a science magazine, a science documentary
 - C** An interactive periodic table, an article published in a science magazine, a science documentary, this Year 11 textbook
 - D** A science article published in a newspaper, an article published in a science magazine, a science documentary, a practical report written by a Year 11
- 15** What is the purpose of referencing and acknowledging documents, ideas and quotations in your investigation?
- 16** For the following source, what is the correct way to make an in-text citation in APA style?
- Selke, S., Auras, R., Nguyen, T.A., Aguirre, E.C., Cheruvathur, R., & Liu, Y. (2015). Evaluation of biodegradation—promoting additives for plastics. *Environmental Science & Technology*, 49(6), 3796–3777.
- A** However, Selke et al. (2015) did not find any significant difference in biodegradability.
 - B** However, Selke et al. did not find any significant difference in biodegradability¹.
 - C** However, Selke et al. did not find any significant difference in biodegradability (Selke, S., Auras, R., Nguyen, T.A., Aguirre, E.C., Cheruvathur, R., & Liu, Y. (2015). Evaluation of biodegradation—promoting additives for plastics. *Environmental Science & Technology*, 49(6), 3769–3777).
 - D** However, Selke et al. (2015) did not find any significant difference in biodegradability (Evaluation of biodegradation-promoting additives for plastics. *Environmental Science & Technology*).
- 17 a** Convert 30.0 mL into L.
b Convert 34.5 kg into g.
- 18** A scientist designed and completed an experiment to test the following hypothesis:
‘Increasing the temperature of water would result in an increase in the measured electrical conductivity of water.’
- a** Write a possible aim for this scientist’s experiment.
 - b** What would be the independent, dependent and controlled variables in this investigation?
 - c** What kind of data would be collected? Would it be qualitative or quantitative?
 - d** List the equipment that could be used and the precision expected for each item.
 - e** Explain the difference between raw data and processed data, using this hypothesis as an example. What would you expect the graph of the results to look like if the scientist’s hypothesis was correct?

APPENDIX A Symbols, units and fundamental constants

TABLE 1 Units and symbols based on the SI system. Units listed in red are the arbitrarily defined fundamental units of the SI system.

Quantity	Symbol for physical quantity	Corresponding SI unit	Symbol for SI unit	Definition of SI unit
Mechanics				
Volume	V	cubic metre	m^3	—
Mass	m	kilogram	kg	fundamental unit
Density	d	—	kg m^{-3}	—
Time	t	second	s	fundamental unit
Force	F	newton	N	kg m s^{-2}
Pressure	P	pascal	Pa	N m^{-2}
Energy	E	joule	J	N m
Electricity				
Electric potential difference	V	volt	V	$\text{J A}^{-1} \text{s}^{-1}$
Nuclear and chemical quantities				
Atomic number	Z	—	—	—
Neutron number	N	—	—	—
Mass number	A	—	—	$Z + N$
Amount of substance	n	mole	mol	fundamental unit
Relative atomic mass	A_r	—	—	—
Relative molecular mass	M_r	—	—	—
Molar mass	M	—	—	kg mol^{-1}
Molar volume	V_m	—	—	$\text{m}^3 \text{mol}^{-1}$
Concentration	c	—	—	mol m^{-3}
Thermal quantities				
Temperature	T	kelvin	K	fundamental unit
Specific heat capacity	C	—	—	$\text{J K}^{-1} \text{kg}^{-1}$

TABLE 2 SI prefixes, their symbols and values

SI prefix	Symbol	Value
pico	p	10^{-12}
nano	n	10^{-9}
micro	μ	10^{-6}
milli	m	10^{-3}
centi	c	10^{-2}
deci	d	10^{-1}
kilo	k	10^3
mega	M	10^6
giga	G	10^9
tera	T	10^{12}

TABLE 3 Some physical constants

Description	Symbol	Value
Avogadro's constant	N_A	$6.022 \times 10^{23} \text{ mol}^{-1}$
Charge of an electron	e	$-1.60 \times 10^{-19} \text{ C}$
Mass of electron	m_e	$9.109 \times 10^{-31} \text{ kg}$
Mass of proton	m_p	$1.673 \times 10^{-27} \text{ kg}$
Mass of neutron	m_n	$1.675 \times 10^{-27} \text{ kg}$
Gas constant	R	$8.31 \text{ J K}^{-1} \text{ mol}^{-1}$
Ionic product for water	K_w	1.00×10^{-14} at 298 K
Molar volume of an ideal gas at 273 K, 100 kPa	V_m	22.71 L mol^{-1}
Specific heat capacity of water	c	$4.18 \text{ J}^{-1} \text{ g}^{-1} \text{ K}^{-1}$
Density of water	d	1.00 g mL^{-1} at 298 K

SIGNIFICANT FIGURES

The number of significant figures a piece of data has indicates the precision of a measurement. For example, compare the following data:

- A jogger takes 20 minutes to cover 4 kilometres.
- A sprinter takes 10.21 seconds to cover 100.0 metres.

The sprinter's data has been measured more precisely than that of the jogger. This is indicated by the greater number of significant figures in the second set of data.

Which figures are significant?

A significant figure is an integer or a zero that follows an integer.

In the data above:

- the distance '4 kilometres' has one significant figure
- the time '20 minutes' has two significant figures (the zero follows the integer 2)
- the 10.21 seconds and 100.0 metres each have four significant figures.

A zero that comes before any integers, however, is not significant. For example:

- the value 0.0004 has only one significant figure, whereas the value 0.0400 has three significant figures. The zeros that come before the integer 4 are not significant, whereas those that follow the integer are significant.

Using significant figures

In chemistry you will often need to calculate a value from a set of data. It is important to remember that the final value you calculate is only as precise as your *least precise piece of data*.

Addition and subtraction

When adding or subtracting values, the answer should have no more digits to the right of the decimal place than the value with the least number of digits to the right of the decimal place.

Example

12.78 mL of water was added to 10.0 mL of water. What is the total volume of water?

$$12.78 \text{ mL} + 10.0 \text{ mL} = 22.78 = 22.8 \text{ mL}$$

Because one of the values (10.0 mL) has only one digit to the right of the decimal place, the answer will need to be adjusted so that it too has only one digit to the right of the decimal place.

Multiplication and division

When multiplying and dividing values, the answer should have no more significant figures than the value with the least number of significant figures.

Example

An athlete takes 3.5 minutes to complete four laps of an oval. What is the average time taken for one lap?

$$\text{Average time} = \frac{3.5 \text{ minutes}}{4} = 0.875 = 0.88 \text{ minutes}$$

Because the data (3.5 minutes) has only two significant figures, the answer will need to be adjusted to two significant figures so that it has the same degree of precision as the data. (Note: The 'four' is taken to indicate a precise number of laps and so is considered to have as many significant figures as the calculation requires. This applies to values that describe *quantities* rather than *measurements*.)

Rounding off

When adjusting the number of significant figures, if the integer after the last significant figure is equal to or greater than '5', then the last significant integer is rounded up. Otherwise, it is rounded down.

STANDARD FORM

A value written in standard form is expressed as a number equal to or greater than 1 and less than 10 multiplied by 10^x , where x is an integer. For example, when written in standard form:

- 360 becomes 3.6×10^2
- 0.360 becomes 3.60×10^{-1}
- 0.000456 becomes 4.56×10^{-5} .

Sometimes you will need to use standard form to indicate the precision of a value.

APPENDIX C Table of relative atomic masses*

TABLE 1 Table of relative atomic masses*

Element name	Symbol	Atomic number	Relative atomic mass	Element name	Symbol	Atomic number	Relative atomic mass	Element name	Symbol	Atomic number	Relative atomic mass
actinium	Ac	89	—	hafnium	Hf	72	178.49	praseodymium	Pr	59	140.9077
aluminium	Al	13	26.9815	hassium	Hs	108	—	promethium	Pm	61	—
americium	Am	95	—	helium	He	2	4.00260	protactinium	Pa	91	231.0359
antimony	Sb	51	121.76	holmium	Ho	67	164.9303	radium	Ra	88	—
argon	Ar	18	39.948	hydrogen	H	1	1.0080	radon	Rn	86	—
arsenic	As	33	74.9216	indium	In	49	114.82	rhenium	Re	75	186.21
astatine	At	85	—	iodine	I	53	126.9045	rhodium	Rh	45	102.9055
barium	Ba	56	137.33	iridium	Ir	77	192.22	roentgenium	Rg	111	—
berkelium	Bk	97	—	iron	Fe	26	55.845	rubidium	Rb	37	85.468
beryllium	Be	4	9.01218	krypton	Kr	36	83.80	ruthenium	Ru	44	101.07
bismuth	Bi	83	208.9804	lanthanum	La	57	138.9055	rutherfordium	Rf	104	—
bohrium	Bh	107	—	lawrencium	Lr	103	—	samarium	Sm	62	150.4
boron	B	5	10.81	lead	Pb	82	207.2	scandium	Sc	21	44.9559
bromine	Br	35	79.904	lithium	Li	3	6.94	seaborgium	Sg	106	—
cadmium	Cd	48	112.41	livermorium	Lv	116	—	selenium	Se	34	78.97
caesium	Cs	55	132.9055	lutetium	Lu	71	174.967	silicon	Si	14	28.086
calcium	Ca	20	40.08	magnesium	Mg	12	24.305	silver	Ag	47	107.868
californium	Cf	98	—	manganese	Mn	25	54.9380	sodium	Na	11	22.9898
carbon	C	6	12.011	meitnerium	Mt	109	—	strontium	Sr	38	87.62
cerium	Ce	58	140.12	mendelevium	Md	101	—	sulfur	S	16	32.06
chlorine	Cl	17	35.453	mercury	Hg	80	200.59	tantalum	Ta	73	180.9479
chromium	Cr	24	51.996	molybdenum	Mo	42	95.95	technetium	Tc	43	—
cobalt	Co	27	58.9332	moscovium	Mc	115	—	tellurium	Te	52	127.60
copernicium	Cn	112	—	neodymium	Nd	60	144.24	tennessine	Ts	117	—
copper	Cu	29	63.55	neon	Ne	10	20.180	terbium	Tb	65	158.9254
curium	Cm	96	—	neptunium	Np	93	—	thallium	Tl	81	204.384
darmstadtium	Ds	110	—	nickel	Ni	28	58.693	thorium	Th	90	232.038
dubnium	Db	105	—	nihonium	Nh	113	—	thulium	Tm	69	168.9342
dysprosium	Dy	66	162.50	niobium	Nb	41	92.9064	tin	Sn	50	118.71
einsteinium	Es	99	—	nitrogen	N	7	14.0067	titanium	Ti	22	47.87
erbium	Er	68	167.26	nobelium	No	102	—	tungsten	W	74	183.84
europium	Eu	63	151.96	oganesson	Og	118	—	uranium	U	92	238.0289
fermium	Fm	100	—	osmium	Os	76	190.2	vanadium	V	23	50.942
flerovium	Fl	114	—	oxygen	O	8	15.9994	xenon	Xe	54	131.29
fluorine	F	9	18.9984	palladium	Pd	46	106.4	ytterbium	Yb	70	173.05
francium	Fr	87	—	phosphorus	P	15	30.9738	yttrium	Y	39	88.9058
gadolinium	Gd	64	157.25	platinum	Pt	78	195.08	zinc	Zn	30	65.38
gallium	Ga	31	69.72	plutonium	Pu	94	—	zirconium	Zr	40	91.22
germanium	Ge	32	72.63	polonium	Po	84	—				
gold	Au	79	196.9666	potassium	K	19	39.098				

* Based on the atomic mass of $^{12}\text{C} = 12$.

The values for relative atomic masses given in the table apply to elements as they exist in nature, without artificial alteration of their isotopic composition, and, further, to natural mixtures that do not include isotopes of radiogenic origin.

Common positive and negative ions and solubility of common ionic compounds in water

TABLE 1 Names and formulae of some common positive and negative ions

Positive ions (cations)						Negative ions (anions)			
+1		+2		+3		-1		-2	
caesium	Cs ⁺	barium	Ba ²⁺	aluminium	Al ³⁺	bromide	Br ⁻	carbonate	CO ₃ ²⁻
copper(I)	Cu ⁺	cadmium(II)	Cd ²⁺	chromium(III)	Cr ³⁺	chloride	Cl ⁻	chromate	CrO ₄ ²⁻
gold(I)	Au ⁺	calcium	Ca ²⁺	gold(III)	Au ³⁺	cyanide	CN ⁻	dichromate	Cr ₂ O ₇ ²⁻
lithium	Li ⁺	cobalt(II)	Co ²⁺	iron(III)	Fe ³⁺	dihydrogen phosphate	H ₂ PO ₄ ⁻	hydrogen phosphate	HPO ₄ ²⁻
potassium	K ⁺	copper(II)	Cu ²⁺			ethanoate	CH ₃ COO ⁻	oxalate	C ₂ O ₄ ²⁻
rubidium	Rb ⁺	iron(II)	Fe ²⁺	+4		fluoride	F ⁻	oxide	O ²⁻
silver	Ag ⁺	lead(II)	Pb ²⁺	lead(IV)	Pb ⁴⁺	hydrogen carbonate	HCO ₃ ⁻	sulfide	S ²⁻
sodium	Na ⁺	magnesium	Mg ²⁺	tin(IV)	Sn ⁴⁺	hydrogen sulfide	HS ⁻	sulfite	SO ₃ ²⁻
		manganese(II)	Mn ²⁺			hydrogen sulfite	HSO ₃ ⁻	sulfate	SO ₄ ²⁻
		mercury(II)	Hg ²⁺			hydrogen sulfate	HSO ₄ ⁻		
		nickel	Ni ²⁺			hydroxide	OH ⁻	-3	
		strontium	Sr ²⁺			iodide	I ⁻	nitride	N ³⁻
		tin(II)	Sn ²⁺			nitrite	NO ₂ ⁻	phosphate	PO ₄ ³⁻
		zinc	Zn ²⁺			nitrate	NO ₃ ⁻		
						permanganate	MnO ₄ ⁻		

TABLE 2 Solubility of common ionic compounds in water

Soluble ionic compounds		
Soluble in water (>0.1 mol dissolves per L at 25°C)	Exceptions: insoluble (<0.01 mol dissolves per L at 25°C)	Exceptions: slightly soluble (0.01–0.1 mol dissolves per L at 25°C)
most chlorides (Cl ⁻), bromides (Br ⁻) and iodides (I ⁻)	AgCl, AgBr, AgI, PbI ₂	PbCl ₂
all nitrates (NO ₃ ⁻)	no exceptions	no exceptions
all ammonium (NH ₄ ⁺) salts	no exceptions	no exceptions
all sodium (Na ⁺) and potassium (K ⁺) salts	no exceptions	no exceptions
all ethanoates (CH ₃ COO ⁻)	no exceptions	no exceptions
most sulfates (SO ₄ ²⁻)	SrSO ₄ , BaSO ₄ , PbSO ₄	CaSO ₄ , Ag ₂ SO ₄
Insoluble ionic compounds		
Insoluble in water	Exceptions: soluble	Exceptions: slightly soluble
most hydroxides (OH ⁻)	NaOH, KOH, Ba(OH) ₂ , NH ₄ OH*, AgOH**	Ca(OH) ₂ , Sr(OH) ₂
most carbonates (CO ₃ ²⁻)	Na ₂ CO ₃ , K ₂ CO ₃ , (NH ₄) ₂ CO ₃	no exceptions
most phosphates (PO ₄ ³⁻)	Na ₃ PO ₄ , K ₃ PO ₄ , (NH ₄) ₃ PO ₄	no exceptions
most sulfides (S ²⁻)	Na ₂ S, K ₂ S, (NH ₄) ₂ S	no exceptions

*NH₄OH does not exist in significant amounts in an ammonia solution. Ammonium and hydroxide ions readily combine to form ammonia and water.

**AgOH readily decomposes to form a precipitate of silver oxide and water.

1. Transforming decimal notation to scientific notation

Scientists use scientific notation to handle very large and very small numbers.

For example, instead of writing 0.000 000 035, scientists would write 3.5×10^{-8} .

A number in *scientific notation* (also called standard form or power of ten notation) is written as:

$$a \times 10^n$$

where a is a number equal to or greater than 1 and less than 10, that is $1 \leq a < 10$

n is an integer (a positive or negative whole number).

n is the power that 10 is raised to and is called the index value.

To transform a very large or very small number into scientific notation:

- 1 write the original number as a decimal number greater than or equal to 1 but less than 10.
- 2 multiply the decimal number by the appropriate power of 10.

The index value is determined by counting the number of places the decimal point needs to be moved to form the original number again.

If the decimal point is moved n places to the right, n will be a positive number.

For example:

$$51 = 5.1 \times 10^1$$

If the decimal point is moved n places to the left, n will be a negative number.

For example:

$$0.51 = 5.1 \times 10^{-1}$$

You will notice from these examples that when large numbers are written in scientific notation, the 10 has a positive index value. When very small numbers are written in scientific notation, the 10 has a negative index value.

Practice questions

- 1 Match each number with its correct scientific notation.

Number	Scientific notation
0.002	2×10^3
2000	1.234×10^{-1}
0.1234	2×10^{-3}
12.34	1.234×10^1
123.4	1.234×10^2

- 2 Write 7.009×10^{-4} using decimal notation.

2. Identifying significant figures

When giving an answer to a calculation it is important to take note of the number of significant figures that you use.

You should give an answer that is as accurate as possible. However, an answer can't be more accurate than the data or the measuring device used to calculate it. For example, if a set of scales that measures to the nearest gram shows that an object has a mass of 56g, then the mass should be recorded as 56g, not 56.0g. This is because you do not know whether it is 56.0g, 56.1g, 56.2g or 55.8g.

56 is a number with 2 significant figures. Recording to 3 significant figures (e.g. 56.0g or 55.8g) would not be scientifically 'honest'.

If this mass of 56 g is used to calculate another value it would also not be 'honest' to give an answer that has more than 2 significant figures.

Determining to what number of significant figures to give an answer to depends on what kind of calculation you are doing.

If you are multiplying or dividing, use the smallest number of significant figures provided in the initial values.

If you are adding or subtracting, use the smallest number of decimal places provided in the initial values.

Working out the number of significant figures

The following rules should be followed to avoid confusion in determining how many significant figures are in a number.

- 1 All non-zero digits are always significant. For example, 21.7 has 3 significant figures.
- 2 All zeroes between two non-zero digits are significant. For example, 3015 has 4 significant figures.
- 3 A zero to the right of a decimal point and following a non-zero digit is significant. For example, 0.5700 has 4 significant figures.
- 4 Any other zero is not significant, as it will be used only for locating decimal places. For example, 0.005 has just 1 significant figure.

Practice questions

- 1 Which of the following is written to 2 significant figures?
 A 30.1
 B 0.000 40
 C 0.5
 D 5.12
- 2 George multiplied 1.22 by 1.364. Which of the options below shows the result of this multiplication with the correct number of significant figures?
 A 1.66
 B 1.664
 C 1.65
 D 1.7
- 3 How can 41 be written to 4 significant figures?
 A 00.41
 B 4100
 C 41.00
 D 4.100
- 4 Alex is getting ready to go for a bike ride. Alex's mass is 65.3 kg. The bicycle has a mass of 12.92 kg.
 - a Calculate the combined mass of Alex and the bicycle. Give your answer to the correct number of significant figures.
 A 78
 B 78.2
 C 78.22
 D 78.3
 - b Using the combined mass calculated in part (a) above, and the formula $F_{\text{net}} = ma$, calculate the force Alex needs to apply to achieve an acceleration of 1.250 ms^{-1} . Give your answer to the correct number of significant figures.

3. Calculating percentages

Scientists use percentages to express a ratio or fraction of a quantity.

To express one quantity as a percentage of another, use the second quantity to represent 100%.

For example, expressing 6 as a percentage of 24 is like saying ‘6 is to 24 as x is to 100’:

$$\begin{aligned}\frac{6}{24} &= \frac{x}{100} \\ x &= \frac{6}{24} \times 100 \\ &= 25\%\end{aligned}$$

To calculate a percentage of a quantity, the percentage is expressed as a decimal then multiplied by the quantity.

For example, to calculate 40% of 20:

$$\begin{aligned}x &= \frac{40}{100} \times 20 \\ &= 0.4 \times 20 \\ &= 8\end{aligned}$$

Practice questions

- What is 9 as a percentage of 12?
A 25%
B 50%
C 75%
D 30%
- What is 25% of 24?
A 6.6
B 6
C 5
D 0.5
- Which of the following values expresses 15 as a percentage of 120?
A 8%
B 5%
C 12.5%
D 0.125%

4. Converting between percentages and fractions

To write a percentage as a fraction, divide the percentage by 100.

For example:

$$\begin{aligned}25\% &= \frac{25}{100} \\ &= \frac{1}{4}\end{aligned}$$

$\frac{25}{100}$ is not the simplest form of this fraction. If you divide both the numerator and the denominator by 25 (their highest common factor) then the fraction simplifies to $\frac{1}{4}$.

Whenever you give a fraction as an answer, always try and simplify it by dividing the numerator and denominator by the highest common factor.

To write a fraction as a percentage, multiply the fraction by 100%. In many cases it is easier to convert the fraction to a decimal number first.

For example:

$$\begin{aligned}\frac{1}{4} &= 0.25 \times 100 \\ &= 25\%\end{aligned}$$

The value of the fraction or percentage has not changed. It is just being represented in a different way.

Practice questions

- Choose the option that expresses $\frac{1}{5}$ as a percentage.
A 25%
B 20%
C 30%
D 50%
- Match each percentage with its corresponding fraction.

Percentage	Fraction
0.2%	$\frac{7}{20}$
2.5%	$\frac{7}{40}$
17.5%	$\frac{1}{40}$
35%	$\frac{111}{250}$
44.4%	$\frac{1}{500}$

5. Changing the subject of an equation

Scientists use equations to represent relationships between variables. In an equation like $A = \pi r^2$, A is called the subject of the equation.

Sometimes the subject of the equation has to be changed in order to express the relationship in a more useful way. For example, if you need to find the radius of a circle, you would want r to be the subject of the equation above.

To change the subject of a simple equation, transpose the equation to leave the new subject on its own. In the example above, the equation needs to read

$$r = \dots$$

Keep the equation balanced by performing the same operation to both sides of the equation to cancel operations being performed on the desired subject. Inverse operations (the opposite operation; for example dividing is the inverse of multiplying) will allow cancelling.

For example, make r the subject of the equation $A = \pi r^2$.

- Divide both sides of the equation by π .

$$A = \pi r^2$$

$$\frac{A}{\pi} = \frac{\pi r^2}{\pi}$$

The π in the numerator and denominator on the right side of the equation cancel out, giving

$$\frac{A}{\pi} = r^2$$

- 2 To cancel the squaring operation of r , take the square root of both sides of the equation.

$$\frac{A}{\pi} = r^2$$

$$\sqrt{\frac{A}{\pi}} = \sqrt{r^2}$$

The square and square root on the right side of the equation cancel out, giving

$$\sqrt{\frac{A}{\pi}} = r$$

- 3 r is now the subject of the equation.

$$r = \sqrt{\frac{A}{\pi}}$$

Practice questions

- 1 Rearrange the formula $A = \frac{2}{3}R$ to make R the subject.

A $R = \frac{2A}{3}$

B $R = \frac{A}{3}$

C $R = \frac{3A}{2}$

D $R = 6A$

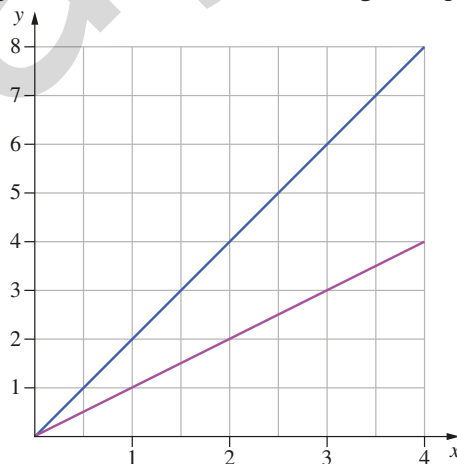
- 2 Rearrange the formula $y = 3\sqrt{\frac{p}{q}}$ to make p the subject.

6. Interpreting the slope of a linear graph

Scientists often represent a relationship between two variables as a graph. For directly proportional relationships, the variables are connected by a straight line, where the slope (or gradient) of the line represents the constant of proportionality between the two variables.

The slope or gradient of the line is defined as the ratio of change between two points in the vertical axis (ΔY), divided by the change between two points in the horizontal axis (ΔX). In other words, it measures the rate at which one variable (the dependent variable) changes with respect to the other (the independent variable).

The graph below has two straight lines with different slopes. The steeper slope (blue line) indicates that the rate of change is higher. This means the change is happening more quickly. On the other hand, the flatter slope (red line) indicates that the rate of change is lower. This means the change is happening more slowly.



Practice questions

- 1 On a graph with two sloped lines, what does the steeper sloped line indicate?
A a faster rate of change
B a slower rate of change
C the same rate of change
D a much slower rate of change
- 2 The rate of change of a straight line on a graph is given by the:
A y-intercept
B x-intercept
C gradient
D area under the graph

7. Understanding mathematical symbols

Part of the language of science is using symbols to represent quantities or to give meanings. For example, the four symbols $<$, $>$, $\leq$ and $\geq$ are known as ‘inequalities’.

The following mathematical symbols are commonly used in science.

Symbol	Meaning	Example	Explanation
$<$	less than	$2 < 3$	2 is less than 3
$>$	greater than	$6 > 1$	6 is greater than 1
$\leq$	less than or equal to	$2x \leq 10$	$2x$ is less than or equal to 10
$\geq$	greater than or equal to	$3y \geq 12$	$3y$ is greater than or equal to 12
$\sqrt{}$	square root	$\sqrt{4} = 2$	The square root of 4 is 2
Δ	change in (difference between)	Δt	change in t (time)
$\approx$	approximately equal to	$\pi \approx 3.14$	π is approximately equal to 3.14
Σ	summation	$\sum_{i=1}^4 i$	The sum of consecutive integers from 1 to 4, i.e. $1 + 2 + 3 + 4 = 10$

Practice questions

- 1 The symbol that means ‘less than’ is:
A $<$
B $>$
C $\leq$
D $\geq$
- 2 Which of these symbols is an inequality?
A $\approx$
B Δ
C $\sqrt{}$
D $\leq$

8. Understanding the difference between discrete and continuous data

Quantitative data forms the backbone of science. Scientists are constantly working with data—measuring, recording, analysing and interpreting it.

Quantitative data consists of numerical values that can either be discrete or continuous.

Discrete data is data that has a set of clearly defined values. For example, the number of students in a class would have a discrete set of possible data.

Continuous data is usually data that is measured in some way and can have an infinite number of values. For example, your height or weight would have a continuous set of possible data.

The easiest way to distinguish between the two types of quantitative data is to ask, 'Is the data measured or counted?' If it is counted, the data set is discrete. If it is measured, the data set is continuous.

Practice questions

- Which one of these data sets is continuous?
A the number of cars parked in a street
B the temperature of the air over a 24-hour period
C the number of students at a school
D the number of nails used to build a fence
- Which one of these data sets is discrete?
A the number of cars parked in a street
B the temperature of the air over 24 hours
C the heights of a team of footballers
D the mass of a team of netballers

9. Calculating the mean, median and range of a data set

When handling data, scientists often look for ways to describe patterns in the data. Common terms used when analysing a set of data include the mean, median and range.

Mean: the *average* value in the data set. To calculate the mean, sum all the values in the data set and then divide this by the number of data values.

Median: the *middle* value in an ordered data set. To calculate the median, arrange the data set in ascending order and then count the number of data values. If the number of values is odd, the median is the middle value. If the number of values is even, calculate the median by adding the two middle values and dividing by 2, i.e. by calculating the average of the two middle numbers.

Range: the *spread* of values in the data set. To calculate the range, take the largest data value and then subtract the smallest data value.

Practice questions

- The following set of data is recorded:
44, 17, 21, 26, 42, 18
Find the:
a mean
b median
c range.
- The mass in kilograms of each student in a class of 25 students is recorded below. The combined mass of all the students is 1340 kg.
Find the:
a mean
b median
c range.
Students' weights: 67, 60, 41, 52, 39, 60, 42, 55, 55, 50, 46, 62, 48, 48, 56, 64, 55, 56, 59, 61, 41, 63, 53, 62, 45

10. Solving simple algebraic equations

To solve an equation means to find the values that make the equation true. Scientists manipulate equations and substitute in known variables in order to solve for the variable required.

For example, you can solve

$$a = \frac{F_{\text{net}}}{m}$$

where F_{net} is the net force on the car, which is 2400 N

m is the mass of the car, which is 1200 kg

a is the acceleration of the car in m s^{-2} , which is unknown.

$$\begin{aligned} a &= \frac{2400}{1200} \\ &= 2 \text{ m s}^{-2} \end{aligned}$$

Practice questions

- Solve the equation $V = IR$ if $I = 3$ and $R = 9$.
A $V = 3$
B $V = 6$
C $V = 12$
D $V = 27$
- Solve the equation and find the value of Q if $Q = mc\Delta T$, $m = 1.2$, $c = 4200$ and $\Delta T = 30$.
A $Q = 840$
B $Q = 25\,200$
C $Q = 126\,000$
D $Q = 6$

11. Completing calculations with more than one operation

Scientists often deal with complex calculations that can involve numerous operations within the one calculation. The order in which these operations are performed can affect the result of the calculation.

For example, the calculation $2 + 3 \times 4$ will give an incorrect answer of 20 if you calculate the $2 + 3$ part first, but it will give the correct answer of 14 if you calculate the 3×4 part first.

Scientists and mathematicians have agreed on a set order in which operations are carried out so that calculations are consistent. You can remember this order using the acronym 'BIDMAS':

- brackets
- indices (powers, square roots, etc.)
- division and multiplication
- addition and subtraction.

The operations present in a calculation are performed in the order shown in the list. If there are multiple instances of division and multiplication, or addition and subtraction, work from left to right.

For example, the 3×4 part in the original example would always be performed first, since multiplication is higher in the list than addition.

When dealing with scientific notation, it is important to keep each individual number complete. Use brackets to help do this, especially when dividing.

For example, when dividing 3.01×10^{21} by 6.02×10^{23} , brackets are used to keep the second number together.

If you used a calculator and entered $\frac{3.01 \times 10^{21}}{6.02 \times 10^{23}}$ (i.e. $3.01 \times 10^{21} \div 6.02 \times 10^{23}$), the answer would come out as 5.00×10^{43} . This is not the correct answer.

The correct answer is only obtained by entering $\frac{3.01 \times 10^{21}}{(6.02 \times 10^{23})}$. This time, the answer comes out correctly as 5.00×10^{-3} .

Using the calculator's *EXP* button or the $\times 10^x$ button keeps the number and power of 10 together as one number and avoids the problems of using the number multiplied by 10^x . If your calculator does not have an *EXP* or $\times 10^x$ button, check the user manual. It may be that it is just labelled differently on your calculator. Alternatively, remember to always use brackets to keep the terms in the denominator together, as shown above.

Practice questions

- What is 3.4×10^{-4} divided by 1.7×10^{-3} ?
A 2×10^{-7}
B 2
C 20
D 0.2
- Substitute $m = 1.4$, $d = 3.9$ and $c = 2.7$ into $W = 6m - 4(d + c)$ and solve for W .
A -4.5
B -18
C 34.8
D 26.7

12. Understanding the relationship between data, graphs and algebraic rules

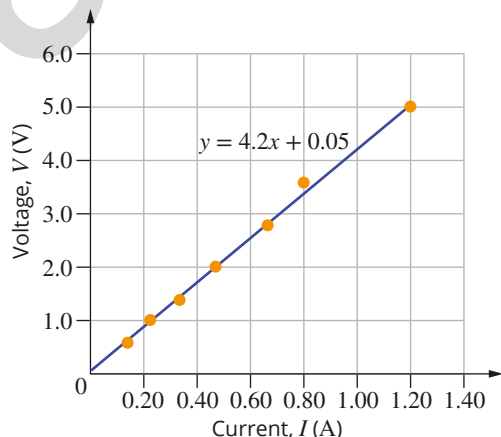
Scientists use graphs to analyse the data they collect from experiments. All graphs tell a story. The shape of the graph shows the relationship between the variables, and this relationship can be written algebraically and numerically. The horizontal axis is known as the x -axis and the vertical axis is known as the y -axis.

Once the algebraic rule is known, the values for one variable can be substituted and the values for the other variable can be calculated. These values can also be determined by reading them from the graph.

For example, when investigating how current and voltage vary across a light bulb, the following data was collected:

Current, I (A)	Voltage, V (V)
0.14	0.6
0.22	1.0
0.33	1.4
0.47	2.0
0.66	2.8
0.80	3.6
1.20	5.0

Graphing this data produced:



The numerical values from the experiment are listed in the table and plotted on the graph. The algebraic relationship between the variables is given by the equation of the line:

$$y = 4.2x + 0.05$$

The value of the y -intercept is approximately zero, so assuming that the y -intercept is zero, and labelling the x -axis as current and the y -axis as voltage, the relationship can be written as:

$$y = 4.2 \times \text{current}$$

Using the appropriate symbols this can also be written as: $V = 4.2I$

Practice questions

- If a graph had L on the y -axis and B on the x -axis and the equation of the straight line was $y = 3.7x$, what is the algebraic form of the graph?
 - $L = 3.7x$
 - $y = 3.7x$
 - $y = 3.7B$
 - $L = 3.7B$
- If a relationship was written as $m = 5.9L$, what shape would the graph be and which variable would be plotted on which axis?
 - The graph would be non-linear, with m on the y -axis and L on the x -axis.
 - The graph would be non-linear, with m on the x -axis and L on the y -axis.
 - The graph would be linear, with m on the y -axis and L on the x -axis.
 - The graph would be linear, with m on the x -axis and L on the y -axis.

13. Recognising and using ratios

A ratio is the relationship between two numbers of the same kind. It could be the quantities in a recipe, the division of profits from a sale, or the number of different types of the same thing.

Scientists use ratios to compare quantities. This might be the numbers of atoms of different elements in a compound, or the number of primary and secondary windings of a transformer.

You can also use the principle of ratios to solve problems. For example, if 1 reaction produces 2.5 MeV of energy, then how much energy does 34 reactions produce?

The reaction-to-energy ratio of 1:2.5 should remain constant as the number of reactions increases. So you need to find the factor that 1 needs to be multiplied by to give 34:

$$1 \times 34 = 34$$

You then multiply the energy amount by the same factor:

$$2.5 \times 34 = 85 \text{ MeV}$$

You may also see ratios expressed as fractions.

Practice questions

- If the primary coil of a transformer has 2400 windings and the secondary coil has 600 windings, what is the simplest ratio of primary to secondary windings?
 - 2400:600
 - 24:6
 - 12:3
 - 4:1

14. Understanding pie charts, frequency graphs, and histograms

It is essential in science to collect data and arrange it in an orderly way. Tables are often used to organise data, which can then be displayed in a graph.

Pie charts

A pie chart is a circle that is divided into sectors. Each sector represents one item in the data set and is shown as a percentage or fraction of the total data set.



Frequency graphs and histograms

Frequency graphs and histograms are another way of representing data visually.

If data is discrete (i.e. can be counted), each column in a column graph will represent one category, e.g. 'apples' or 'strawberries'. Often these columns have a gap between them.

If the data is continuous (i.e. can be measured), such as the heights of the students in that class, each column will represent a range of possible heights, e.g. 140 to 160 cm, and there will be no gaps between the columns. These are called histograms.

Practice questions

- 1 Choose the pie chart that correctly shows the data from Emmanuel's poll of soccer fans.

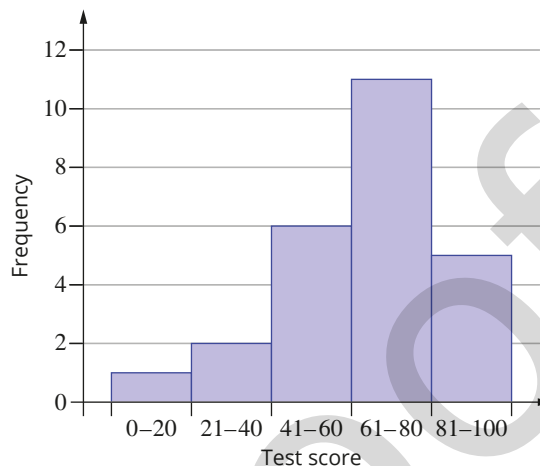
Number of matches watched	10	12	15	18	21
Relative frequency (%)	25	35	15	15	10

Matches watched:

-  = 10
-  = 18
-  = 12
-  = 21
-  = 15



- 2 A class of Year 11 Chemistry students had a test. A histogram of their test scores is shown below.



- How many students scored over 80?
- How many students scored below 40?
- How many student are in this class?

15. Using units in an equation to check for dimensional consistency

Scientists know that each term in an equation represents a quantity. The units used to measure that quantity are not used in the calculations. Units are only indicated on the final line of the solved equation.

For example, this is the equation for the area (A) of a rectangle of length (L) and width (W):

$$A = L \times W$$

If L has a value of 7 m and W has a value of 4 m, it is written:

$$\begin{aligned} A &= L \times W \\ &= 7 \times 4 \\ &= 28\text{m}^2 \end{aligned}$$

Note that the units are left out of the actual calculation on the second line and only included at the end, after the numerical answer.

You can use units to check the dimensional consistency of the answer. In the example above, the two quantities of L (length) and W (width) both have to be expressed in consistent units, in this case metres (m), to give an answer that is expressed in square metres ($\text{m} \times \text{m} = \text{m}^2$).

If you had made a mistake, and used the formula $A = L + W$ instead, the answer would be expressed in metres only. This is not the correct unit to express area, so you would know that was wrong.

Practice questions

- Which formula has the correct dimensions for calculating volume in m^3 ?
 - $\text{m} \times \text{m}$
 - $\text{m} \times \text{s}$
 - $\text{m} \times \text{m} \times \text{s}$
 - $\text{m} \times \text{m} \times \text{m}$

- 2 Which of these shows the correct substitution into $p = 2L + 2W$ using consistent units for $L = 3.5 \text{ cm}$ and $W = 240 \text{ cm}$?
- A $P = 2 \times 3.5 + 2 \times 240$
 B $P = 2 \times 3.5 + 2 \times 24$
 C $P = 2 \times 35 + 2 \times 240$
 D $P = 2 \times 3.5 + 2 \times 2.4$
- 3 By using the equation $p = mv$ as a guide, select the correct units in which to measure momentum, p .
- A ms
 B ms^{-1}
 C kg ms^{-1}
 D kg ms^{-2}

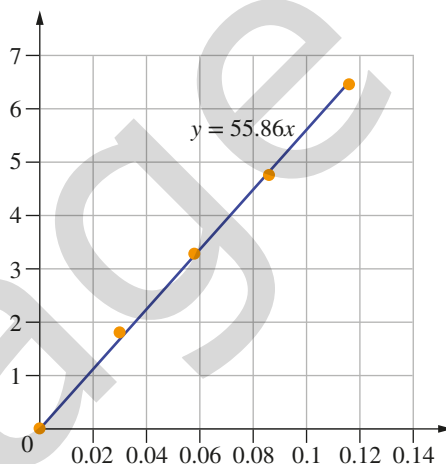
16. Understanding lines of best fit

When scientists observe that the points on a graph seem to form a straight line, then a line of best fit (or trendline) can be drawn through them. Computer programs such as Excel can fit a trendline to a data set; lines of best fit can also be drawn by hand onto a printed or drawn graph.

The line of best fit should pass as close as possible to as many of the points as possible (i.e. it should 'fit' the data closely). It may not pass exactly through any of the points, but once the line of best fit is drawn, the points should be spaced equally on each side, above and below the line. There should be no points very far away from the line, unless they are considered to be unreliable. Unreliable points are called 'outliers' and can be disregarded for the purposes of creating a line of best fit.

The gradient and y -intercept of the line (not the points) can be determined to find the relationship between the variables using the general equation for a straight line: $y = mx + c$.

For example:



Practice questions

- 1 Are the following statements true or false?
- All the points on a scatter plot must lie on the line of best fit.
 - The line of best fit may pass through none of the points.
 - The points of the scatter plot should lie close to the line of best fit.
- 2 Are the following statements true or false?
- A trendline is the same as a line of best fit.
 - A line of best fit can only be drawn using a computer program such as Excel.
 - Outliers should be included as normal points when considering where to draw a line of best fit.

Answers to practice questions

Appendix E Maths Skills

1. Transforming decimal notation to scientific notation

1	0.002	2×10^{-3}
	2000	2×10^3
	0.1234	1.234×10^{-1}
	12.34	1.234×10^1
	123.4	1.234×10^2

0.002: Move the decimal point 3 places to the right which gives an index of -3 , so 0.002 is written as 2×10^{-3} .

2000: Move the decimal point 3 places to the left which gives an index of $+3$, so 2000 is written as 2×10^3 .

0.1234: Move the decimal point 1 place to the right which gives an index of -1 , so 0.1234 is written as 1.234×10^{-1} .

12.34: Move the decimal point 1 place to the left which gives an index of $+1$, so 12.34 is written as 1.234×10^1 .

123.4: Move the decimal point 2 places to the left which gives an index of $+2$, so 123.4 is written as 1.234×10^2 .

- 2 The -4 index shows the decimal point has been moved 4 places to the right. Move it back 4 places to the left to give 0.0007009.

2. Identifying significant figures

- 1 B 0.000 40

- 2 A 1.66

When two numbers are multiplied, use the smallest number of significant figures in the initial values to give your answer. In George's multiplication, the answer is 1.66408, but as 1.22 has 3 significant figures, the correct answer is 1.66.

- 3 C 41.00

The zeroes to the right of the decimal point are significant. When writing an answer to a correct number of significant figures, you may need to use rounding if the initial answer has more figures (digits) than you need. These extra figures are called 'non-significant figures.' If the first non-significant figure is ≥ 5 , you round up. If the first non-significant figure is < 5 , then do not round up. For example, if the initial answer for a calculation was 2.1259 but you only needed an answer to 3 significant figures, you would round it to 2.13. If the initial answer was 2.1241, then to 3 significant figures this becomes 2.12.

- 4 a B 78.2

When 65.3 is added to 12.92, the answer is 78.22, but only if it is assumed that 65.3 is actually 65.30. When adding or subtracting with significant figures, use the smallest number of significant figures provided in the initial values. As there is no way of knowing the accuracy of beyond 65.3, the answer should only be given to 3 significant figures, which is 1 decimal place, 78.2.

- b When multiplying or dividing, use the smallest number of significant figures in the initial values to give your answer.

$$F_{\text{net}} = ma$$

$$= 78.2 \times 1.250$$

$$= 97.8$$

Since the mass value only has 3 significant figures, the answer should also only have 3 significant figures. Therefore, even though the calculated answer is 97.75, the answer to 3 significant figures is 97.8 ms^{-1} .

3. Calculating percentages

- 1 C 75%

$$\frac{9}{12} = \frac{x}{100}$$

$$x = \frac{9}{12} \times 100$$

$$= 75\%$$

- 2 B 6

$$x = \frac{25}{100} \times 24$$

$$= 0.25 \times 24$$

$$= 6$$

- 3 C 12.5%

$$\frac{15}{120} = \frac{x}{100}$$

$$x = \frac{15}{120} \times 100$$

$$= 12.5\%$$

4. Converting between percentages and fractions

- 1 B 20%

To write a fraction as a percentage, multiply the fraction by 100.

$$\frac{1}{5} = 0.2 \times 100$$

$$= 20\%$$

- 2

0.2%	$\frac{1}{500}$
2.5%	$\frac{1}{40}$
17.5%	$\frac{7}{40}$
35%	$\frac{7}{20}$
44.4%	$\frac{111}{250}$

To change a percentage to a fraction, divide by 100.

$$0.2\% = \frac{0.2}{100} = \frac{2}{1000} = \frac{1}{500}$$

$$2.5\% = \frac{2.5}{100} = \frac{25}{1000} = \frac{1}{40}$$

$$17.5\% = \frac{17.5}{100} = \frac{175}{1000} = \frac{35}{200} = \frac{7}{40}$$

$$35\% = \frac{35}{100} = \frac{7}{20}$$

$$44.4\% = \frac{44.4}{100} = \frac{444}{1000} = \frac{222}{500} = \frac{111}{250}$$

5. Changing the subject of an equation

1 C

$$R = \frac{3A}{2}$$

Multiply both sides of the equation by 3

$$3A = 3 \times \frac{2}{3}R$$
$$= 2R$$

Divide both sides of the equation by 2

$$\frac{3A}{2} = R$$

Rewrite the equation so it reads

$$R = \frac{3A}{2}$$

2 $p = \frac{y^2q}{9}$

Divide both sides of the equation by 3

$$\frac{y}{3} = \sqrt{\frac{p}{q}}$$

Square both sides

$$\left(\frac{y}{3}\right)^2 = \frac{p}{q}$$

Expand the brackets on the left

$$\frac{y^2}{9} = \frac{p}{q}$$

Multiply both sides by q

$$\frac{y^2q}{9} = p$$

Rewrite the equation so it reads

$$p = \frac{y^2q}{9}$$

6. Interpreting the slope of a linear graph

- 1 A. The steepness of the slope indicates the rate of change. A line with a steeper slope indicates a faster rate of change.
- 2 C. The gradient or slope of a linear graph indicates the rate of change.

7. Understanding mathematical symbols

- 1 A <. One way is to remember that the smaller end of the shape points toward the smaller number. For example, $3 < 6$ means 3 is less than 6.
- 2 D ≤. The symbols <, >, ≤ and ≥ are all inequalities.

8. Understanding the difference between discrete and continuous data

- 1 B the temperature of the air over a 24-hour period.
If the data can be counted, the data set is discrete. If the data can be measured, the data set is continuous. The temperature of the air can be measured with a thermometer, so the data set is continuous.
- 2 A the number of cars parked in a street.
If the data can be counted, the data set is discrete. If the data can be measured, the data set is continuous. The number of cars parked in a street can be counted, so the data set is discrete.

9. Calculating the mean, median and range of a data set

- 1 a $44 + 17 + 21 + 26 + 42 + 18 = 168$
 $\frac{168}{6} = 28$
b Place the numbers in ascending order: 17, 18, 21, 26, 42, 44
As there is an even number of values, add the two middle values and divide by 2
 $\frac{21+26}{2} = 23.5$
c $44 - 17 = 27$
- 2 a $\frac{1340}{25} = 53.6$ kg
b 55 kg
c $67 - 39 = 28$ kg

10. Solving simple algebraic equations

- 1 D $V = 27$
Substitute the values and solve the equation.
 $V = 3 \times 9$
 $= 27$
- 2 B $Q = 25\,200$
Substitute the values and solve the equation.
 $Q = 0.2 \times 4\,200 \times 30$
 $= 25\,200$

11. Completing calculations with more than one operation

- 1 D 0.2
The correct calculation, using brackets, is:
 $\frac{3.4 \times 10^{-4}}{(1.7 \times 10^{-3})} = 0.2$
- 2 B -18
 $W = 6 \times 1.4 - 4 \times (3.9 + 2.7)$
 $= 6 \times 1.4 - 4 \times 6.6$
 $= 8.4 - 26.4$
 $= -18$

12. Understanding the relationship between data, graphs and algebraic rules

- 1 D $L = 3.7B$
Substituting L for y and B for x gives $L = 3.7B$.
- 2 C. The graph would be linear, with m on the y -axis and L on the x -axis.
The graph would be linear as the equation is written in the form $y = mx + c$ with m on the y -axis and L on the x -axis.

13. Recognising and using ratios

- 1 D 4:1
The primary to secondary ratio is 2400:600. Dividing both sides of the ratio by 600 simplifies it to 4:1.

14. Understanding pie charts, frequency graphs, and histograms

1 A



The largest sector of the pie chart is purple, representing the percentage of people who watched 12 matches (35%). The next in size is blue (10 matches, 25%), then green (18 matches, 15%) and yellow (15 matches, 15%). The smallest sector is red (21 matches, 10%).

- 2 a 5
b 3
c 25
- The height of the 81–100 column is 5. This shows 5 students scored over 80.
The heights of the 0–20 and 21–40 columns are 1 and 2.
This shows that 3 students scored below 40.
If you add up the heights of all the columns ($1 + 2 + 6 + 11 + 5 = 25$), it tells you that there are 25 students in the class.

15. Using units in an equation to check for dimensional consistency

- 1 D $m \times m \times m$
2 D $P = 2 \times 3.5 + 2 \times 2.4$
- Both units must be consistent.
Since $100\text{ cm} = 1\text{ m}$ the consistent units must be either
 $L = 3.5\text{ m}$ and $W = 240\text{ cm}$
or
 $L = 350\text{ cm}$ and $W = 240\text{ cm}$
The only correct combination is $P = 2 \times 3.5 + 2 \times 2.4$.

- 3 C kgms^{-1}
- The unit for momentum is taken from the unit for mass (kg) multiplied by the unit for velocity (ms^{-1}). Therefore, momentum is measured in kgms^{-1} .

16. Understanding lines of best fit

- 1 a False.
b True.
c True.
2 a True.
b False.
c False.

Glossary

A

absorbance A measure of the capacity of a substance to absorb light of a specified wavelength.

accurate If the average of a set of measurements of a quantity is very close to the true or accepted value of the quantity, then the measurement is described as accurate.

acid A substance capable of producing hydrogen ions in solution (Arrhenius model) or donating a hydrogen ion (Bronsted-Lowry model).

acid rain Rainwater that has reacted with acidic emissions and has a pH less than 5.5.

acid-base reaction A reaction in which an acid reacts with a base.

acidic proton A proton bonded to an electronegative element (oxygen, nitrogen or fluorine) that is donated to a base during an acid-base reaction.

acidic solution An aqueous solution in which the concentration of hydronium ions (H^+) is greater than the concentration of hydroxide ions (OH^-). At 25°C , $\text{pH} < 7$.

acidification An increase in atmospheric carbon dioxide concentration causing an increase in the acidity of the oceans.

acidity The concentration of H^+ ions in an aqueous solution. Acidity is measured using the pH scale.

activation energy The minimum energy required by reactants for a reaction to occur; symbol E_a . This energy is needed to break the bonds between atoms in the reactants to allow products to form.

addition reaction A reaction in which a molecule binds to an unsaturated hydrocarbon, forming a single carbon-carbon bond. In this process two reactant molecules form one product.

alkali A soluble base or a solution of a soluble base.

alkali metal A group 1 metal—Li, Na, K, Rb, Cs and Fr.

alkane A saturated hydrocarbon; general formula $\text{C}_n\text{H}_{2n+2}$.

alkene An unsaturated hydrocarbon containing one carbon-carbon double bond; general formula C_nH_{2n} .

alkyl group A group obtained by removing a hydrogen atom from an alkane; general formula $\text{C}_n\text{H}_{2n+1}$, e.g. methyl ($-\text{CH}_3$).

allotrope Different forms of the same element in which the atoms combine in different ways.

alloy A substance formed when other materials (e.g. carbon, other metals) are mixed with a metal.

alpha particle A positively charged particle formed from ionising radiation.

amorphous A structure that has no consistent arrangement of particles.

amount A measure used by chemists for counting particles; the unit is the mole.

analyte The chemical substance that is of interest in a chemical analysis.

anion A negatively charged ion, e.g. a chloride ion, Cl^- .

annealing Heating a metal to a moderate temperature and then allowing it to cool slowly to make it softer and more ductile.

aqueous When a chemical species has been dissolved in water, the resulting solution is said to be aqueous. This can be shown by writing '(aq)' after the name or symbol of the chemical.

Arrhenius model A model that defines an acid as a substance that ionises water to produce H^+ ions and a base as a substance that dissociates in water to form OH^- ions.

artesian basin An underground area of porous rock surrounded by rock that is not permeable to water. Rain seeps into the rock and is stored underground.

asymmetrical molecule A molecule in which the polar bonds are unevenly (or asymmetrically) distributed. The bond dipoles do not cancel and an overall molecular dipole is created.

atom The basic building block of matter. It is made up of subatomic particles—protons, neutrons and electrons.

atomic absorption spectroscopy (AAS) An analytical technique that uses light absorption to measure the concentration of a metal in a sample.

atomic number The number of protons in the nucleus of an atom; identical to the charge number of the nucleus; symbol Z .

atomic radius A measurement used for the size of atoms; determined by measuring half the distance between two adjacent atoms in a molecule or structure.

atomic theory of matter A theory proposed by John Dalton in 1802 that states that all matter is made up of atoms. He said that atoms are indivisible, atoms of the same element are identical and compounds are made up of different types of atoms in fixed ratios.

Avogadro's law At the same temperature and pressure, equal volumes of all gases contain equal numbers of particles.

Avogadro's number The number of particles in a mole; symbol $N_A = 6.022 \times 10^{23} \text{ mol}^{-1}$.

B

ball-and-stick model A model that displays both the three-dimensional position of the atoms and the bonds between them. The atoms are represented by coloured spheres that are connected by rods to represent the bonds.

base A substance capable of producing hydroxide ions in solution (Arrhenius model) or accepting a proton (Bronsted-Lowry model).

basic solution An aqueous solution in which $[\text{H}^+] < [\text{OH}^-]$. For a basic solution at 25°C , $\text{pH} > 7$.

bias Measured values consistently in one direction from the actual value; they may be too high or too low.

bibliography A list of resources referred to in your research.

biodegradable Capable of being decomposed by bacteria or other living organisms.

biodiesel A renewable fuel usually produced from vegetable oil from sources such as soyabean, canola or palm oil.

bioethanol A renewable fuel produced from crops such as sugar cane.

biofuel Fuel derived from plant materials such as grains (maize, wheat, barley or sorghum), sugar cane and vegetable waste, and vegetable oils.

biogas A renewable fuel that can be used to generate electricity.

Bohr model A theory of the atom proposed by Niels Bohr that states that electrons in an atom occupy fixed, circular orbits that correspond to specific energy levels.

bore water Water collected in aquifers (underground water-bearing rock) below Earth's surface. Bore water may be accessed by drilling and sinking a bore pipe into the aquifer.

brittle Shatters when given a sharp tap.

Bronsted-Lowry theory A theory that defines an acid as a proton (hydrogen ion) donor and a base as a proton acceptor. In the reaction $\text{HCl(g)} + \text{H}_2\text{O(l)} \rightarrow \text{H}_3\text{O}^+(\text{aq}) + \text{Cl}^-(\text{aq})$, HCl is the proton donor and is classified as an acid. H_2O is a proton acceptor and is classified as a base.

buckyball A ball-like polyhedral molecule consisting of carbon atoms of the type found in fullerenes.

C

calibrate To determine, check or rectify the graduation of any instrument giving quantitative measurements.

calibration curve A plot of data involving two variables that is used to determine the values for one of the variables.

carbon-12 The isotope of carbon that has a mass number of 12. The isotope contains 6 protons and 6 neutrons. One atom of carbon-12 is taken as having a mass of exactly 12 atomic mass units. This is the standard from which all other relative masses are calculated.

carbon nanotube A tube-shaped nanoparticle, made of carbon atoms.

carbon neutral A description of a process that overall does not contribute to a change in atmospheric carbon.

carrier gas An unreactive gas, often nitrogen, used in gas chromatography.

catalyst A substance that increases the rate of a reaction but is not consumed in the reaction. The catalyst provides a new reaction pathway with a lower activation energy.

cation A positively charged ion, e.g. a sodium ion, Na^+ .

centrifugation A separation process that uses the action of rapid rotation to accelerate the settling of particles in a solid-liquid mixture.

ceramic Material that is produced by the firing (heating followed by cooling) of clay.

chemical code Classification of chemicals into hazardous categories.

chemical contaminant An element or a compound that may be harmful if consumed in drinking water. It can be naturally occurring or synthetic.

chemical energy Form of energy stored in the chemical bonds between atoms and molecules. This energy results from: attractions between electrons and protons in atoms; repulsions between nuclei; repulsions between electrons; movement of electrons; vibrations and rotations around bonds.

chemical formula A representation of a substance using symbols for its constituent elements. It shows the ratio of atoms present in the substance.

chemical symbol A symbolic representation of an element, usually one or two letters, where the first letter is capitalised and the second letter is lower case, e.g. carbon's symbol is C and sodium's symbol is Na.

chemiluminescence The light that emitted by a chemical reaction that does not produce significant amounts of heat.

chemiluminescent Emitting light as the result of a chemical reaction.

chlorination A step in the purification of drinking water. The clear water is treated with gaseous chlorine to destroy any bacteria; the main purpose of chlorination is to remove biological contaminants.

chromatogram The output of a chromatography procedure. In TLC and paper chromatography, it is the pattern of bands or spots formed on a plate or on the paper. In HPLC, it is the graph produced.

chromatography A technique for separating the components of a mixture. The components are carried by a mobile phase over the adsorbent surface of the stationary phase.

collision theory A theoretical model that accounts for rates of chemical reactions in terms of collisions between particles occurring during a chemical reaction.

colloid A mixture in which very small particles are spread throughout a liquid, solid or gas. The particles are bigger than single molecules, but so small that they do not settle on standing.

column chromatography A chromatographic technique in which the stationary phase is in a column, e.g. high performance liquid chromatography (HPLC).

combustion A rapid reaction with oxygen accompanied by the release of large amounts of heat; also called burning.

complete combustion A hydrocarbon undergoes complete combustion with oxygen at high temperatures when the only products are carbon dioxide and water.

components The chemicals in a mixture.

composite material A combination of two or more distinct materials with significantly different physical and chemical properties.

compound A pure substance made up of different types of atoms combined in a fixed ratio.

concentrated solution A solution that has a relatively high ratio of solute to solvent.

concentration A measure of how much solute is dissolved in a specified volume or mass of solution.

condensed structural formula A simple representation of the structural formula of an organic molecule. A condensed structural formula shows the atoms connected to each carbon atom, but not all the bonds, e.g. $\text{CH}_3\text{CH}_2\text{CH}_3$.

conductivity Permitting the flow of electric charges.

conductor An object or type of material that permits the flow of electric charges, e.g. a wire is an electrical conductor that can carry electricity along its length.

control experiment An identical experiment carried out at the same time, but the independent variable is not changed.

controlled variable A variable that must be kept constant during an investigation.

core charge (or effective nuclear charge) The effective nuclear charge experienced by the outer-shell electrons in an atom. It indicates the attractive force felt by the valence electrons towards the nucleus. The effective nuclear charge experienced by the outer-shell electrons in an atom. It indicates the attractive force felt by the valence electrons towards the nucleus.

covalent bond The force of attraction formed when one or more pairs of electrons are shared between two nuclei.

covalent layer lattice An arrangement of atoms in a lattice in which there are strong covalent bonds between the atoms that have formed in a layer.

covalent network lattice An arrangement of atoms in a lattice in which there are strong covalent bonds between the atoms in all three dimensions.

cracking A chemical process during which carbon-carbon bonds in alkanes are broken to form smaller molecules and some unsaturated molecules.

credible Reliable and can be backed up with evidence; a credible source provides information that one can believe to be true.

crude oil A mixture of hydrocarbons that originates from the remains of prehistoric marine micro-organisms. The organisms have been broken down by high temperatures and pressures over millions of years; also called petroleum.

crystal A solid made up of atoms or molecules arranged in a repeating three-dimensional pattern.

crystal lattice The symmetrical three-dimensional arrangement of atoms or ions inside a crystal.

crystallisation The process in which solid crystals are deposited when the concentration of a solute in a solution increases past the point of saturation.

crystallise Form solid crystals.

D

decantation A separation process in which a layer of liquid is separated from solid that has settled at the bottom of the liquid.

decomposition A reaction in which a compound is broken down into smaller parts.

delocalise Spread out.

delocalised electron An electron that is not restricted to the region between two atoms.

density A measure of the amount of mass per unit volume. It has the SI units of kg m^{-3} , but is commonly quoted in g cm^{-3} or mL^{-1} .

dependent variable The variable that is measured or observed to determine the effect of changes in the independent variable.

desalinated seawater Fresh water made by removing the salt from seawater.

desalination The removal of salts from seawater to obtain fresh water.

desorption The breaking of the attraction between a substance and the surface to which the substance is adsorbed.

diamond A form of pure carbon that is the hardest naturally occurring substance.

diatomic molecule A molecule formed from two atoms only, e.g. Cl_2 .

diffusion All the gases in a gas mixture spread to fill the available space. The speed at which the gases spread depends upon the mass of the gas particles.

dilute solution A solution that has a relatively low ratio of solute to solvent.

dilution The addition of a solvent to a solution to reduce its concentration.

dipole The separation of areas of positive and negative charge in a molecule.

dipole-dipole force A form of intermolecular force that occurs between polar molecules where the partially positively charged end of one molecule is attracted to the partially negatively charged end of another molecule.

diprotic acid An acid that can ionise in water to form two H^+ ions.

discrete A separate object or particle.

dispersion force The force of attraction between molecules due to temporary dipoles induced in the molecules. The temporary dipoles are the result of fluctuations in the electron density.

dissociate Break up.

dissociation A process in which molecules or ionic compounds separate or split into smaller particles such as atoms or ions. Examples of dissociation reactions include the solution of NaCl solid in water, forming $\text{Na}^+(\text{aq})$ and $\text{Cl}^-(\text{aq})$ ions, and the reaction of HCl gas with water, forming $\text{H}^+(\text{aq})$ and $\text{Cl}^-(\text{aq})$.

dissolution The process of dissolving a solute in a solvent to form a solution.

dissolve To incorporate a solid or gas into a liquid so as to form a solution.

distillation The process of separating a solution by evaporating, condensing and collecting the component of the solution with the lowest boiling point.

double covalent bond A covalent bond in which four electrons (two electron pairs) are shared.

ductile Able to be drawn into a wire.

E

electric dipole see *dipole*

electrical conductivity The degree to which a specified material or solution conducts an electric current.

electrolyte A solution or molten substance that conducts electricity by means of the movement of ions, e.g. a solution of sodium chloride ($\text{NaCl}(\text{aq})$).

electromagnetic radiation A form of energy that moves through space. Visible light, radio waves and X-rays are forms of electromagnetic radiation.

electromagnetic spectrum All possible frequencies of electromagnetic radiation shown in order of their wavelengths or frequencies.

electron A negatively charged, subatomic particle that occupies the region around the nucleus of an atom.

electron configuration In the shell model of an atom, the electronic configuration is a means of representing the number of electrons in each shell.

electron density The concentration of electrons that usually refer to the regions around an atom or molecule.

electron dot diagram See *Lewis structure diagram*.

electron shell In the shell model of an atom, an electron shell is the fixed energy level that corresponds to a circular orbit of the electrons. In the Schrödinger model, a shell contains subshells and orbitals of equal or similar energy.

electron transfer diagram A diagram that shows how electrons move from a metal atom to a non-metal to form ions.

electronegativity The ability of an atom to attract electrons in a covalent bond towards itself.

electrostatic attraction The force of attraction between a positively charged particle and a negatively charged particle.

element A substance made up of atoms with the same atomic number.

elemental analysis A process that determines the mass of each element in a sample of a compound.

eluent The solvent that carries the components and passes through a chromatography column.

emission line When an electron absorbs energy it jumps to a higher energy state. Shortly afterwards the electron returns to the lower energy level, releasing a fixed amount of energy as a particular colour of light.

emission spectrum A spectrum produced when an element is excited by heat or radiation. It appears as distinct lines characteristic of the element.

empirical formula A formula that shows the simplest whole number ratio of the elements in a compound, e.g. CH_2 is the empirical formula of propene (C_3H_6).

endothermic A reaction that absorbs energy from the surroundings; ΔH is positive.

energy content The chemical energy of a substance.

energy density The energy released per litre of fuel.

energy efficiency The percentage of energy from a source that is converted to useful energy

energy level One of the different shells of an atom in which an electron can be found.

enhanced greenhouse effect Increasing levels of greenhouse gases produced by our use of fossil fuels are causing global warming and triggering consequential shifts in weather patterns and climate.

enhanced greenhouse effect Increasing levels of greenhouse gases produced by our use of fossil fuels are causing global warming and triggering consequential shifts in weather patterns and climate.

enthalpy Heat content. The sum of the chemical potential and kinetic energies in a substance; symbol H .

enthalpy change The difference in the total enthalpy of the products and the total energy of the reactants; symbol ΔH . Also known as heat of reaction.

enzyme Biological catalyst.

excess reactant A reactant that is not completely consumed in a chemical reaction.

excited electron An electron that has absorbed a particular quantity of energy and moved from its ground state to a higher energy level. An atom is said to be excited when an electron is not in the lowest electron energy level possible (i.e. not in the ground state).

excited state A term used to describe an atom in which electrons occupy higher energy levels than the lowest possible energy levels.

expertise Expert knowledge or skills in a field.

F

fermentation A process which uses other enzymes from yeasts to convert glucose and other small sugar molecules to ethanol and carbon dioxide.

filtrate The purified liquid that collects in the flask when a mixture is filtered.

filtration The process of removing solids from a liquid or gas by passing the mixture through a fine mesh or filter.

first ionisation energy The energy required to remove one electron from an atom of an element in the gas phase.

flame test Determination of the metallic elements present in a compound by inserting a sample of the compound into a non-luminous Bunsen burner flame. Some metals produce particular colours when they are heated.

floc Aluminium hydroxide precipitate produced in the flocculation process, which traps other fine particles and removes colour and some microorganisms from the water.

flocculation Process in water purification where small suspended particles come together to form larger, heavier particles, which are usually insoluble.

fluoridation Process in which fluoride is added to drinking water before it is released from storages. Fluoride in drinking water helps to reduce the incidence of tooth decay.

formula A representation of an element or compound using symbols for its constituent elements. It shows the ratio of atoms in a compound or the number of atoms in a molecule.

fossil fuel A fuel formed by the decomposition of plant and animal material over millions of years, including coal, oil and natural gas.

fracking Process of extraction of natural gas from coal or shale deposits.

fractional distillation A form of distillation that separates solutions of liquids according to their boiling point.

full equation A representation of a reaction that uses the formulae of all reactants and products in a chemical reaction.

fullerene A molecule composed entirely of carbon, in the form of a hollow sphere or tube. Each carbon atom is bonded to three other carbon atoms.

G

gas chromatography A very sensitive form of chromatography in which the mobile phase is a gas.

geometric isomer A type of stereoisomer that can occur when there is restricted rotation in a molecule. Restricted rotation can occur about a carbon-carbon double bond or a ring. Stereoisomers of alkenes can be distinguished using the labels *cis*- and *trans*-.

giant molecule A molecule formed by some non-metals that contains a huge number of atoms, e.g. the large network lattices such as diamond and graphite.

graphene A form of carbon consisting of planar sheets one atom thick in which each carbon atom is bonded to three neighbouring carbon atoms.

graphite A form of carbon in which the carbon atoms are arranged in layers.

Great Artesian Basin A very large reservoir of groundwater in Australia. It is the largest basin of its kind in the world.

greenhouse effect The warming of the Earth's atmosphere due to the absorption of infrared radiation by gases such as carbon dioxide, water and methane.

greenhouse gas Gases in the atmosphere that absorb and re-radiate the energy as infra-red radiation. Carbon dioxide, methane, water vapour, nitrogen oxides and ozone are greenhouse gases.

ground state A term used to describe an atom in which the electrons occupy the lowest possible energy levels.

groundwater Water found below the Earth's surface in porous rock or fractures.

group (periodic table) A vertical column of elements in the periodic table.

H

heat capacity A measure of a substance's capacity to absorb and store heat energy. The heat capacity of 1 g of water is $4.18 \text{ J}^\circ\text{C}^{-1}$. This tells you 1 g of water will absorb 4.18 J of heat energy to heat up by 1°C .

heat of combustion The enthalpy change that occurs when a specified amount (e.g. 1 g, 1 L, 1 mol) of the fuel burns completely in oxygen.

heat of reaction The exchange of heat between a system and its surroundings during a chemical reaction under constant pressure; symbol ΔH . Also known as enthalpy change.

heat treatment Heating a metal in different ways to alter its structure and physical properties.

heavy metal A metal with high density; usually used to describe a metal that poses a threat to health.

heterogeneous Diverse, different. A heterogeneous substance or solution possesses two or more different types of phases in the one sample, e.g. a suspension.

heterogeneous catalyst A catalyst that has a different physical state (phase) from the reactants and products.

high-performance (or high-pressure) liquid chromatography (HPLC) A very sensitive technique used to separate the components in a mixture, to identify each component, and to measure the concentrations of the components. It relies on pumps to pass a pressurised liquid solvent containing the sample mixture through a column filled with a solid adsorbent material.

homogeneous Uniform. The components of a homogeneous substance are uniformly distributed throughout the substance, e.g. a solution is homogeneous because the solute and the solvent cannot be distinguished from each other.

homogeneous catalyst A catalyst that has the same physical state (phase) as the reactants and products.

homologous series A series of compounds with similar properties and the same general formula, in which each member contains one CH_2 unit more than the previous member.

hydrated An ion surrounded by water molecules. Hydrated ions can be found in aqueous solutions or crystalline solids.

hydride A compound in which hydrogen is bonded to another element. HF , HCl and HI are hydrides of group 17 elements.

hydrocarbon A compound that contains carbon and hydrogen only, e.g. the alkanes, alkenes and alkynes.

hydrogen bond A type of intermolecular, dipole-dipole force where a hydrogen atom is covalently bonded to a highly electronegative atom such as oxygen, nitrogen or fluorine. Due to the disparity of electronegativity values between the atoms involved, the hydrogen develops a partial positive charge and bonds to lone pairs of electrons on neighbouring atoms of oxygen, nitrogen or fluorine.

hydronium ion The $\text{H}_3\text{O}^+(\text{aq})$ ion.

hydroxide ion The $\text{OH}^-(\text{aq})$ ion.

hypothesis A prediction based on previous knowledge; a possible outcome of the experiment.

I

ideal gas A gas that obeys the gas equation at all temperatures and pressures. An ideal gas is a theoretical gas in which there are no intermolecular attractive forces and the gas particles have no volume.

incomplete combustion Combustion that takes place when the oxygen supply is limited. Products are carbon monoxide (or carbon) and water.

independent variable A variable that is changed by the researcher.

indicator A substance that is different colours in its acid and base forms.

inert Not chemically reactive.

injection port Opening through which the sample is injected in gas chromatography.

insecticide A substance used for killing insects.

instantaneous dipole A net dipole formed in a molecule due to temporary fluctuations in the electron density in the molecule.

intermolecular force An electrostatic force of attraction between molecules, including dipole-dipole forces, hydrogen bonds and dispersion forces.

interstitial alloy An alloy made by adding smaller atoms to a metal.

intramolecular bond A force that holds the atoms within a molecule together.

ion A positively or negatively charged atom or group of atoms.

ion-dipole attraction The attraction that forms between dissociated ions and polar water molecules when an ionic solid dissolves in water.

ionic bonding A type of chemical bonding that involves the electrostatic attraction between oppositely charged ions.

ionic compound A type of chemical compound that involves the electrostatic attraction between oppositely charged ions.

ionic equation An equation for a reaction that shows only the species involved in the reaction, e.g. $\text{Cu}^{2+}(\text{aq}) + 2\text{OH}^{-}(\text{aq}) \rightarrow \text{Cu}(\text{OH})_2(\text{s})$.

ionisation (i) The removal of one or more electrons from an atom or ion; (ii) the reaction of a molecular substance with a solvent to form ions in solution.

ionisation energy The energy required to remove one electron from an atom of an element in the gas phase.

ionise The reaction of a molecular substance with a solvent to form ions in solution. When some polar molecules dissolve in water they ionise to form a hydrogen ion, e.g. $\text{HCl}(\text{g}) \rightarrow \text{H}^{+}(\text{aq}) + \text{Cl}^{-}(\text{aq})$.

isotope Each of two or more forms of the same element that contain equal numbers of protons but different numbers of neutrons in their nuclei, e.g. ^{12}C and ^{13}C are isotopes of carbon.

K

kinetic energy The energy that a particle or body has due to its motion ($\text{KE} = \frac{1}{2}mv^2$).

kinetic energy distribution diagram See *Maxwell-Boltzmann distribution curve*.

kinetic molecular theory A theory that aims to explain the behaviour of gases by assuming gases are composed of a large number of particles in random motion, these particles move in straight lines and have elastic collisions, the gas particles are very small and there is no attraction between the particles and the average kinetic energy is related to the temperature of the gas.

L

latent heat The heat energy required to change the state of a substance at a constant temperature.

latent heat of fusion The energy required to change a fixed amount of solid to liquid at its melting temperature. The latent heat of fusion of water is 6.0 kJ mol^{-1} , meaning 6.0 kJ of energy is needed to change one mole of water from a solid to a liquid at 0°C .

latent heat of vaporisation The energy required to change a fixed amount of liquid to a gas at its boiling temperature. The latent heat of vaporisation of water is 44.0 kJ mol^{-1} , meaning 44.0 kJ is needed to change one mole of water from a liquid to a gas at 100°C .

lattice A regular arrangement of large numbers of atoms, ions or molecules.

law of conservation of energy A scientific law that states energy cannot be created or destroyed; it can only be transformed from type of energy to another.

Lewis structure diagram A representation of the electron arrangement in a molecule in which outer-shell electrons are represented by dots or crosses; also called electron dot diagram.

limewater test A test for carbon dioxide gas. The presence of carbon dioxide is detected by bubbling the gas through a calcium hydroxide solution ($\text{Ca}(\text{OH})_2(\text{aq})$). The limewater reacts with the carbon dioxide and turns milky.

limiting reactant A reactant that is completely consumed in a reaction and which determines the amount of products formed.

line of best fit See *trend line*.

line spectrum See *emission spectrum*.

liquefied petroleum gas (LPG) A fuel comprising of a mixture of propane and butane, which are liquefied under pressure.

lone electron pair An outer-shell electron pair that does not form a bond with other atoms.

lone pair See *lone electron pair*.

M

main group element An element in groups 1, 2 or 13–18 in the periodic table.

malleable Able to be bent or beaten into sheets.

mass number The number of protons and neutrons in the nucleus of an atom.

mass spectrometer An instrument that measures the mass-to-charge ratio of particles.

mass spectrum A plot of the isotopic mass, relative to the mass of carbon-12 taken as 12 units exactly, against the relative abundance of each isotope present in a sample.

matter Anything that has mass and occupies space.

Maxwell-Boltzmann distribution curve A graph showing the range of kinetic energies of particles in a sample; also called a kinetic energy distribution diagram.

mean The sum of all the values in a data set divided by the number of values in the data set. It is commonly known as the average of a set of numbers.

metallic bonding The electrostatic attractive forces between delocalised valence electrons and positively charged metal ions.

metallic bonding model A description that explains the properties and behaviour of metals in terms of the particles in metals.

metallic character Describes how closely an element exhibits the properties commonly associated with metals, namely, that it readily loses an electron to form a cation. This is closely related to ionisation energy.

metallic lattice Tightly packed arrangement of metal atoms in a 'sea' of electrons.

metallic nanomaterial A material with metal atoms arranged to make structures which have a least one dimension between 1–100 nm ($1\text{ nm} = 10^{-9}\text{ m}$).

metalloid An element that displays both metallic and non-metallic properties, e.g. germanium, silicon, arsenic, tellurium.

mineral A naturally occurring inorganic substance that is solid and can be represented by a chemical formula, e.g. quartz.

miscible Liquids that can be mixed in any ratio to form a homogeneous solution.

mobile phase The phase that moves over the stationary phase in a chromatographic separation.

model A description that scientists use to represent the important features of a system or phenomenon.

molar mass The mass of one mole of a substance measured in g mol^{-1} ; symbol M .

molarity The amount of solute, in moles, dissolved in 1 litre of solution (mol L^{-1}).

molar volume The volume occupied by one mole of gas at a specified set of conditions. At standard temperature (0°C or 273 K) and pressure (100 kPa), the molar volume of a gas is 22.7 L mol^{-1} .

mole The amount of substance that contains the same number of fundamental particles as there are atoms in 12 g of carbon-12; symbol n ; unit mol.

mole ratio The ratio of species involved in a chemical reaction, based on the ratio of their coefficients in the reaction equation.

molecular formula A formula of a compound that gives the actual number and type of atoms present in a molecule. It may be the same as or different from the empirical formula.

molecule A group of two or more atoms covalently bonded together.

molten Materials that are normally found as solids but are liquid, melted, due to elevated temperature.

monatomic ion An ion that is made up of only one element, e.g. neon gas.

monoprotic acid An acid molecule that generates only one hydrogen ion when ionised in water.

N

nanomaterial A material with nanoscale features.

nanometre 10^{-9} of a metre.

nanoparticle A particle in the size range 1–100 nm.

nanorod A nanoscale rod in which length and width are in the range 1–100 nm. The length of the rod is 3–5 times its width.

nanoscale The scale used to classify objects 1–100 nm in size.

nanotechnology The use of technologies that manipulate and investigate the properties of nanomaterials.

nanotube An allotrope of carbon that consists of layers of carbon atoms formed into a long cylinder.

nanowire A wire that has a diameter measured on the nanoscale. Its length is unrestricted.

natural gas A fossil fuel mainly composed of methane (CH_4) together with small amounts of other hydrocarbons such as ethane (C_2H_6) and propane (C_3H_8). Water, sulfur, carbon dioxide and nitrogen may also be present in natural gas.

neutral solution A solution in which the concentrations of H^+ ions equals the concentration of OH^- ions; is neither acidic nor basic. At 25°C a neutral solution has a pH of 7.

neutralisation reaction An acid reacts with a base in stoichiometric proportions to form a salt plus water.

neutralise To react an acid with a base in stoichiometric proportions to form a solution of a salt and water.

neutron An uncharged subatomic particle found in the nucleus of an atom.

noble gas An unreactive gaseous element in group 18 of the periodic table. With the exception of helium, noble gases have eight electrons in their outer shells.

non-bonding electron An outer-shell electron that is not shared between atoms.

non-polar Bonds or molecules that do not have a permanent dipole. They have an even distribution of charge.

non-polar solvent A liquid or solvent that is a compound of two or more elements whose electronegativities are almost the same, e.g. oil.

non-renewable Resources that are used faster than they can be replaced, e.g. coal, oil and natural gas.

nucleon A particle that makes up the nucleus of an atom, i.e. protons and neutrons.

nucleus The positively charged core at the centre of an atom, consisting of protons and neutrons.

O

octet rule A rule used as part of the explanation for electron configuration and in bonding. The rule is that during a chemical reaction, atoms tend to lose, gain or share their valence electrons so that there are eight electrons in the outer shell.

ore A mineral or an aggregate of minerals that contains a valuable constituent, such as a metal, which is mined or extracted.

origin The point at which a small spot of a mixture is placed so that it can be separated by paper or thin-layer chromatography.

osmosis The tendency of a solvent to pass through a semipermeable membrane from an area of low solute concentration to an area of higher solute concentration.

outermost shell See *valence shell*.

outlier A value that lies outside most of the other values in a set of data.

P

paper chromatography An analytical technique for separating and identifying mixtures which uses paper as the stationary phase.

partial pressure The pressure exerted by one component of a mixture of gases. The total pressure of a mixture of gases is equal to the sum of the individual partial pressures of each component in the mixture.

parts per million (ppm) A unit of concentration which states the number of grams of solute in 1 million grams of solution. It is equivalent to the number of mg of solute per kg of solution.

pascal Unit of pressure equal to 1 newton per square metre (1 N m^{-2}), where: pressure = $\frac{\text{force}}{\text{area}}$.

percentage composition The proportion by mass of the different elements in a compound.
$$\% \text{ by mass of an element in a compound} = \frac{\text{mass of the element present}}{\text{total mass of the compound}} \times 100$$

period (periodic table) A horizontal row of elements in the periodic table. The start of a new period corresponds to the outer electron of that element beginning a new shell.

periodic law The way properties of elements vary periodically with their atomic number.

periodic table A table that organises the elements by grouping them according to their electronic configurations.

petrodiesel A hydrocarbon fraction obtained from crude oil.

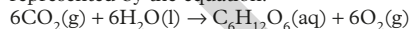
pH A measure of acidity and the concentration of hydronium ions, in solution. Acidic solutions have a pH value less than 7 at 25°C and bases have a pH value greater than 7 (at 25°C). Mathematically, pH is defined as $\text{pH} = -\log_{10}[\text{H}^+]$.

pH scale See *pH*.

phenomenon Something that occurs and can be observed (or felt, heard etc.) to have occurred.

photochemical smog Atmospheric pollution produced through the action of sunlight on nitrogen oxides and unburned hydrocarbons to form ozone and other pollutants. The nitrogen oxides formed in high temperature reactions such as those that occur in car engines and lightning strikes.

photosynthesis A reaction that occurs in the leaves of plants between carbon dioxide and water, in the presence of sunlight and chlorophyll, to form glucose and oxygen. The photosynthesis process can be represented by the equation:



photovoltaic cell A device constructed from a specialised semiconductor that can produce a flow of electrons from light energy.

plastic A property of a material that can be reshaped by application of heat and pressure. In society, some polymers are often referred to as plastics.

polar Bonds or molecules with a permanent dipole. They have an uneven distribution of charge.

polar solvent A solvent that can dissolve substances consisting of polar molecules or charged ions, but will not dissolve solutes made up of non-polar molecules. For example, the polar nature of the ethanol molecule means it readily dissolves in water, which is also polar.

polarity The measure of how polar a molecule or bond is. The difference in charge between the positive and negative ends of an electric dipole. The difference in charge between the positive and negative ends of a polar molecule or covalent bond.

polyatomic ion An ion that is made up of more than one element, e.g. the carbonate ion (CO_3^{2-}).

polyatomic molecule A molecule that consists of more than two types of atoms, e.g. H_2O .

polymer A long-chain molecule that is formed by the reaction of large numbers of repeating units (monomers).

polyprotic acid An acid molecule that generates more than one hydrogen ion when ionised in water.

potable water Water that is suitable for drinking.

precipitate The solid formed during a reaction in which two or more solutions are mixed.

precipitation reaction A reaction between substances in solution in which one of the products is insoluble.

precision When repeated measurements of the same quantity give values that are in close agreement.

pressure The force exerted per unit area of a surface.

primary source A source that is a first-hand account.

protected catchment An area set aside to harvest and store water. There is no public access to protected catchments to ensure the quality of the water is high.

proton A positively charged, subatomic particle bound to neutrons in the nucleus of an atom.

Q

qualitative Relating to quality and not measured values.

qualitative analysis An analysis to determine the identity of the chemical(s) present in a substance.

quantised In specific quantities or chunks.

quantitative Relating to measured values.

quantitative analysis An analysis to determine the concentration of the chemicals present in a mixture.

quenching Heating a metal to a moderate temperature and then cooling it rapidly to make it harder and more brittle.

R

radioactive Spontaneously undergoing nuclear decay to produce radiation such as beta particles, alpha particles and gamma rays.

random error An error that is caused by measurements of a quantity being either above or below the actual quantity being measured. (The effects of random errors can be reduced by taking the average of many observations.)

range The spread of values in the data set. It is taken as the largest data value minus the smallest data value.

rate of reaction The change in concentration of a reactant or product per unit time.

raw data The information and results collected and recorded during an experiment.

reaction pathway A series of chemical reactions that converts a starting material into a product in a number of steps.

reactivity The ease with which a chemical can undergo reactions.

reactivity series of metals A ranking of metals in increasing order of their reactivity (ability to be oxidised) with the half-equations written as reduction equations of the corresponding ion. Least reactive metals are at the top and most reactive metals are at the bottom.

recycled water Water recovered by the purification of wastewater.

relative atomic mass The weighted average of the relative isotopic masses of an element on the scale where ^{12}C is 12 units exactly; symbol A_r .

relative formula mass The mass of a formula unit relative to the mass of an atom of ^{12}C taken as 12 units exactly. It is numerically equal to the sum of the relative atomic masses of the atoms making up the formula. Substances that contain atoms or ions bonded in lattice structures have a relative formula mass. Such compounds include ionic compounds and covalent network substances.

relative isotopic abundance The percentage abundance of a particular isotope in a sample of an element.

relative isotopic mass The mass of an atom of the isotope relative to the mass of an atom of carbon-12 taken as 12 units exactly.

relative molecular mass The mass of a molecule relative to the mass of an atom of ^{12}C , taken 12 units exactly; symbol M_r .

reliability A measure of how an experiment can be repeated several times with consistent results.

renewable Energy that can be obtained from natural resources that can be constantly replenished.

reputation The belief that someone or something has a particular characteristic.

research question A statement defining what is being investigated.

retardation factor (R_f) The ratio of the distance a component has moved from the origin to the distance the solvent has moved from the origin.

retention time (R_t) The time taken for a component to pass through a chromatography column.

reverse osmosis A process by which pure water can be obtained from salt water. Pressure is applied to the salt water, causing a net flow of water molecules away from the solution through a semipermeable membrane.

risk assessment A formal way of identifying risks and assessing potential harm from hazards in an experiment.

S

Safety Data Sheet (SDS) A summary of the risks of using a particular chemical, including measures to be followed to reduce risk.

salinity The presence of salt in water and soil that can damage plants or inhibit their growth.

salt A substance formed from a metal or ammonium cation and an anion. Salts are the products of reactions between acids and bases, metal oxides, carbonates and reactive metals.

saturated (i) A hydrocarbon that is composed of molecules with only carbon-carbon single bonds. (ii) Combined with or containing all the solute that can normally be dissolved at a particular temperature.

saturated solution A solution that cannot dissolve any more solute at the given temperature.

scanning tunnelling microscope (STM) A microscope that images atoms by using a sharp metallic tip to sense the atoms on the surface of a crystal.

secondary source A source derived from the original data or account.

sediment The solid material that settles to the bottom of a body of water.

sedimentation The process of solid particles settling out of a liquid.

seed crystal A small crystal from which a large crystal of the same material can typically be grown.

separation funnel A funnel with a large bulb and a tap in its output tube that is used to separate immiscible liquids.

settling A process in the purification of drinking water that utilises gravity to remove suspended solids from the water.

sieving The process of separating large solid particles from smaller solid particles by passing the mixture through a mesh.

SI units International System of Units based on the metre, kilogram, second, ampere, kelvin, candela and mole.

side group A group of atoms attached to a backbone chain of a long molecule.

significant figures The numbers that convey meaning and precision; usually depends on the scale of the measuring instrument.

single covalent bond A covalent bond in which two electrons are shared between two nuclei. It is depicted in a valence structure as a line between the two atoms involved.

solubility A measure of the amount of solute dissolved in a given amount of solvent at a given temperature.

solubility curve A graph of solubility versus temperature for a particular solute dissolved in a particular solvent.

solubility table A reference table that can be used to predict the solubility of ionic compounds.

solute A substance that dissolves in a solvent, e.g. sugar is the solute when it dissolves in water.

solution A homogeneous mixture of a solute dissolved in a solvent.

solvent A substance, usually a liquid, which is able to dissolve a solute to form a solution. Water is a very good solvent.

space-filling model A three-dimensional model in which the atoms are represented by spheres with radii proportional to the radii of the atoms involved. The distances between spheres are also proportional to the distances between the atomic nuclei.

specific heat capacity The amount of energy required to raise the temperature of an amount of a substance, usually 1 gram, by 1°C . The unit for specific heat capacity is usually $\text{J g}^{-1}^\circ\text{C}^{-1}$, e.g. the specific heat capacity of water is $4.18 \text{ J g}^{-1}^\circ\text{C}^{-1}$.

spectator ion An ion that remains in solution and is unchanged in the course of a reaction. Spectator ions are not included in ionic equations.

spectroscopy The study of the way that radiation, such as light and radio waves, interacts with matter.

standard solution A solution that has an accurately known concentration.

standard temperature and pressure (STP) Conditions of temperature and pressure relevant to a gas, where the temperature is 0°C (273 K) and pressure is 100 kPa .

stationary phase A solid, or a solid that is coated in a viscous liquid, used in chromatography. The components of a mixture undergo adsorption to this phase as they are carried along by the mobile phase.

steel A generally hard, strong, durable and not malleable alloy of iron and carbon, usually containing 0.2–1.5% carbon, often with other constituents such as manganese, chromium, nickel, molybdenum, copper, tungsten, cobalt or silicon, depending on the desired alloy properties.

stem name The name that corresponds to the prefix for the longest chain of carbons in the molecule.

sterile An environment that is totally clean and free from living microorganisms.

stoichiometric calculation The calculation of relative amounts of reactants and products in a chemical reaction.

stoichiometry The calculation of relative amounts of reactants and products in a chemical reaction. Chemical equations give the ratios of the amounts (moles) of the reactants and products.

strong acid An acid that completely ionises in aqueous solution.

strong base A base that completely dissociates in aqueous solution.

structural formula A formula that represents the three-dimensional arrangement of atoms in a molecule and shows all bonds as well as all atoms.

structural isomer A compound with the same molecular formula, but different structures.

subatomic particle A particle that makes up an atom—protons, neutrons and electrons.

sublimation The process by which a substance goes directly from the solid phase to the gaseous phase, without passing through a liquid phase.

substitutional alloy An alloy made from elements of similar chemical properties and size.

successive ionisation energy The energy required to achieve the sequential removal of electrons from the atom.

supersaturated solution An unstable solution that has more solute dissolved at a given temperature than a saturated solution.

surface area (chemistry) The area of all surfaces of the substance that are exposed to the other reactants. This is proportional to the amount of particles available at the surface to react.

surface tension The resistance of a liquid to increase its surface area.

surroundings (chemistry) The rest of the universe around a particular chemical reaction. The chemical reaction is the system. Energy moves between the system and surroundings in exothermic and endothermic reactions.

suspension A heterogeneous mixture containing solid particles that are large enough to sink to the bottom of the mixture if it is left to stand (sedimentation), e.g. sand in a container of water that has just been shaken.

sustainability The combination of meeting the long-term on going needs of society, whilst still reaching the immediate and short-term demands.

symmetrical molecule A molecule in which the polar bonds are evenly (or symmetrically) distributed. The bond dipoles cancel out and do not create an overall molecular dipole.

system In chemistry, the system is a chemical reaction. A system operates within its surroundings. Energy can move between the two. You always consider energy as absorbed or released from the perspective of the system. For example, energy is absorbed into the system from the surroundings, or energy is released by the system to the surroundings.

systematic error An error that produces a constant bias (measurements are either always above, or always below, the actual value) in measurement. (Systematic errors are eliminated or minimised through calibration of apparatus and the careful design of a procedure.)

T

tempering A process in which a metal that has been quenched is warmed again to a lower temperature to reduce its brittleness but to retain its hardness.

temporary dipole A net dipole formed in a molecule due to temporary fluctuations in the electron density in the molecule.

tensile strength The maximum resistance of a material to a force which is pulling it apart before breaking, measured as the maximum stress the material can withstand without tearing.

tetrahedral shape The shape of a molecule with a central atom surrounded by four other atoms. The bond angle between two outer atoms and the central atom is 109.5° .

thermochemical equation A chemical equation that includes the enthalpy change of the reaction, ΔH . For example, $2\text{H}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{H}_2\text{O}(\text{g})$ $\Delta H = -572 \text{ kJ mol}^{-1}$.

thin-layer chromatography An analytical technique for separating and identifying mixtures; it uses a thin layer of fine powder spread on a glass or plastic plate as the stationary phase.

transition metal An element in groups 3–12 in the periodic table.

transition state An arrangement of atoms in a reaction that occurs when sufficient energy is absorbed for the activation energy to be reached. It represents the stage of maximum potential energy for the reaction. Bond breaking and bond forming are both occurring at this stage, and the arrangement of atoms is unstable.

trend An observed pattern of data in a particular direction.

trend line Points can be joined with a single smooth straight or curved line; also known as a line of best fit.

triple covalent bond A covalent bond in which six electrons are shared between two nuclei. It is depicted in a valence structure as three lines between the two atoms involved.

triprotic acid An acid molecule that generates three hydronium ions when ionised in water.

U

uncertainty An error associated with measurements made during experimental work.

unsaturated A hydrocarbon composed of molecules with one or more carbon–carbon double or triple bonds.

unsaturated solution A solution that could dissolve more solute at the given temperature.

V

valence electron An electron found in the valence shell; an outermost electron in an atom or ion.

valence shell The highest energy shell (outer shell) of an atom that contains electrons.

valence shell electron pair repulsion theory (VSEPR) A model used to predict the shape of molecules. The basis of VSEPR is that the valence electron pairs surrounding an atom mutually repel each other, and therefore adopt an arrangement that minimises this repulsion, thus determining the molecular shape.

valence structure A representation of the electron arrangement in a molecule in which covalent bonds are represented by lines.

valency The charge of an ion is equal to its valency. It is a measure of the combining ability of an atom with other atoms.

validity Refers to whether the evidence supports the argument.

vapour pressure The pressure exerted by molecules that evaporate from a liquid in an enclosed vessel at the point where the rates of evaporation and condensation are in equilibrium.

variable Any factor that can be controlled, changed and measured in an experiment.

volume The amount of space that a substance occupies. It can be calculated by multiplying length by width by depth of a regular solid. Otherwise it can be determined by finding the volume of water displaced by the substance.

W

water cycle The continuous process by which water is circulated throughout the Earth and the atmosphere. Water passes into the atmosphere as water vapour, precipitates to Earth in liquid or solid form, and returns to the atmosphere through evaporation.

weak acid An acid that is partly ionised in water.

weak base A base that partially dissociates in water to produce hydroxide (OH^-) ions.

work hardening The increase in strength that results from processes such as hammering, rolling and drawing a metal.